

SOFTWARE ARCHITECTURE AND DESIGN

Project Overview and Software Requirement Specification

FreshMart: Online Fruit Shopping Platform

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Record of Changes (*A - Added, M - Modified, D - Deleted)

Date	A* M, D	In charge	Change Description
5/23/2026	A, M	HungDN	Modify Project Information, add 1.3 Product Background
5/23/2026	A	QuangTM	Add Context Diagram
5/25/2026	A	LamTB	Add ERD, add ERD description
5/25/2026	A	LamTB	Add guest use case diagram
5/25/2026	A	QuangTM	Add Usecase list, Actors,UC Specifications for UC01,02,03,04,05,06,07
5/27/2026	A	LamTB	Add Usecase list, Actors,UC Specifications for UC14,15,16,17,18,24,32,33,34,35,55, BR for each usecase
5/27/2026	A	AnhDD	Add UCs Diagram for shipper, UCs Specifications for UC 44, 45, 46, 56
5/27/2026	A	HungDN	Add Use case Specifications for UC08
5/28/2026	A, M	HungDN	Add, Modify Use case Specifications for UC09,10,11,12,13,19,20,21,22,23
5/28/2026	M	HungDN	Modify the entire of Use case Specifications
5/28/2026	A,M	HungDN	Add, Modify 4.Non-Function Requirements
5/28/2026	M	HungDN	Add 5. Requirement Appendix and informations of them
5/29/2026	A	HuyNQ	Add Feature tree and Feature list
5/29/2026	A	HuyNQ	Add UC Specification for UC 25 26 27 28 29 30 31 36 37 38 39 40 41 42 43
5/29/2026	A	AnhDD	Add UCs Specifications for UC 47, 48, 49, 50, 51, 52, 53, 54
5/29/2026	A	QuangTM	Add main flow 1.2.1, 1.2.2 swimlane
5/30/2026	A	LamTB	Add Screen Flow And Screen Authoring , main flow 1.2.3, 1.2.4 swimlane
5/30/2026	A,M	HungDN	Add main flows 1.2.5, 1.2.6 swimlane; Modify 5.1 Business rules, Modify BR from 3. Use case Specifications

5/30/2026	A, M	HungDN	Modify feature tree and feature list, Add UseCase Specification : UC-57,58,59,60,61,62,63,64
5/30/2026	A	AnhDD	Add main flows 1.2.8, 1.2.9 swimlane
5/30/2026	A	HuyNQ	Add main flow 1.2.7, 1.2.10 swimlane
6/02/2026	M	QuangTM	Update context diagram, UC Customer
6/3/2026	M	HungDN, AnhDD	Modify feature tree and feature list, feature from UCs List
6/3/2026	M	LamTB	Modify screen flow, feature list

I. Project Overview

1.1. Project Information

- English name: FreshMart - Online Fruit Shopping Platform
- Vietnamese name: Nền tảng mua sắm trái cây trực tuyến
- Team name: SE28 NJ - Team 5
- Software type: Website

1.2. Project Team

StudentID	Full Name	Role	Email
HE191209	Đỗ Nguyễn Hùng	Leader	hungdonguyen123@gmail.com
HE191359	Trần Bảo Lâm	Member	lamtranbao1234@gmail.com
HE191413	Đỗ Đức Anh	Member	doducanh03012005@gmail.com
HE187247	Nguyễn Quang Huy	Member	ngquanghuy.work@gmail.com
HE186215	Trịnh Minh Quang	Member	quangtmhe186215@fpt.edu.vn

1.3. Product Background

Customers who purchase fruits from traditional markets or physical stores often spend significant time traveling, selecting products, and completing payments. In many cases, they may not find the desired products, lack sufficient information about fruit quality, or face difficulties in comparing prices and availability across different vendors. These limitations make traditional fruit shopping inconvenient, particularly for office workers, students, and busy families who require faster and more flexible purchasing options.

On the other hand, fruit shop owners and managers also face challenges in manually managing products, inventory, and customer orders. Tasks such as tracking stock levels, identifying low-stock or expired items, processing orders, and analyzing sales performance can become inefficient and error-prone when handled without a centralized system.

Therefore, **FreshMart – the Online Fruit Shopping Platform** is proposed as a multi-vendor e-commerce system to address these issues. The platform aims to provide customers with a convenient online shopping experience while enabling managers, shippers, and administrators to efficiently manage products, orders, deliveries, and system operations in a centralized and automated manner.

1.4. Major Features

FE-01: Authentication & User Management

Allow users to register, log in, log out, recover passwords, and manage personal profile information including password updates.

FE-02: Shop Management

Allow users to browse shops, view shop details, and administrators to create, update, delete, and assign shops to managers for multi-shop operations.

FE-03: Product Management

Allow customers to browse products across shops, search and filter products, view product details, while managers manage product CRUD and product status within assigned shops.

FE-04: Shopping Cart Management

Allow customers to add products from multiple shops, manage cart items, update quantities, and remove products.

FE-05: Checkout & Payment Processing

Allow customers to place orders grouped by shop, select delivery address, shipping method, and process payments via COD or online payment gateways.

FE-06: Order Management & Customer Purchase Lifecycle

Allow customers to track orders, view order details, cancel eligible orders, submit product reviews, and request refunds or returns.

FE-07: Delivery & Shipment Management

Allow shippers and managers to manage deliveries including assignment, status updates, delivery proof, and failure handling.

FE-08: Inventory Management

Allow managers to manage stock intake, update inventory quantities, and monitor stock levels and expiration status within their assigned shops.

FE-09: Category Management

Allow administrators to manage global product categories used across all shops in the platform, ensuring consistent product classification.

FE-10: Analytics & Reporting

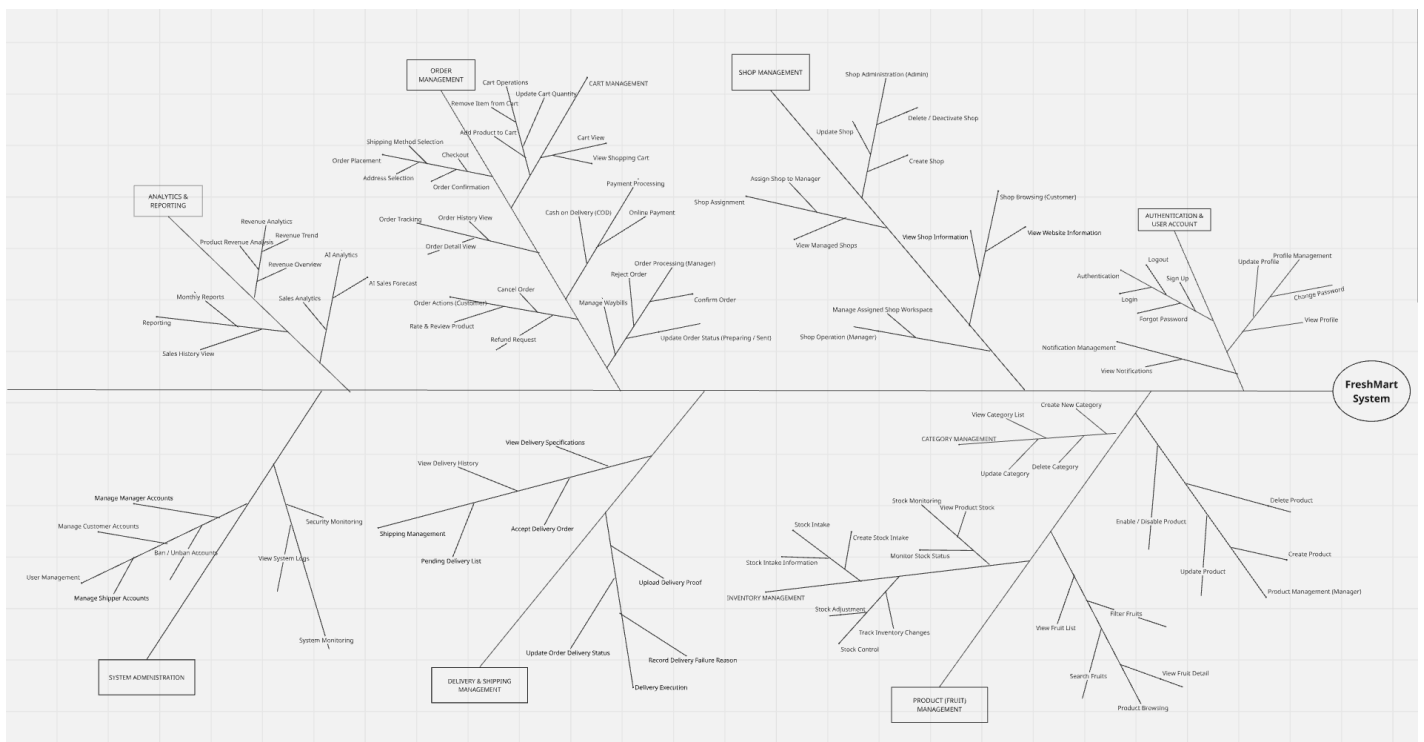
Allow managers to view revenue analytics, sales performance, product performance, and business reports across one or multiple assigned shops.

FE-11: AI Forecasting & Predictive Analytics

Provide AI-powered forecasting for revenue trends, demand prediction, and inventory optimization based on historical multi-shop data.

FE-12: System Administration & Monitoring

Allow administrators to manage users, roles, permissions, system logs, security monitoring, and automated system reports.



[Feature tree](#)

1.5. Tools & Infrastructures

Category	Tools / Infrastructure
Technology	React 18 (FrontEnd), Spring Boot (Java) (BackEnd)
Database	My SQL
API	Restful API
IDEs/Editors	Visual Studio Code, IntelliJ IDEA
Diagramming	Visual paradigm

Documentation	Google Docs, Google Sheets
Version Control	Git
Deployment server	Local Server
Test tool	Postman, Swagger
Project management	GitLab, Excel

II. Software Requirement Specification

1. Overall Requirement

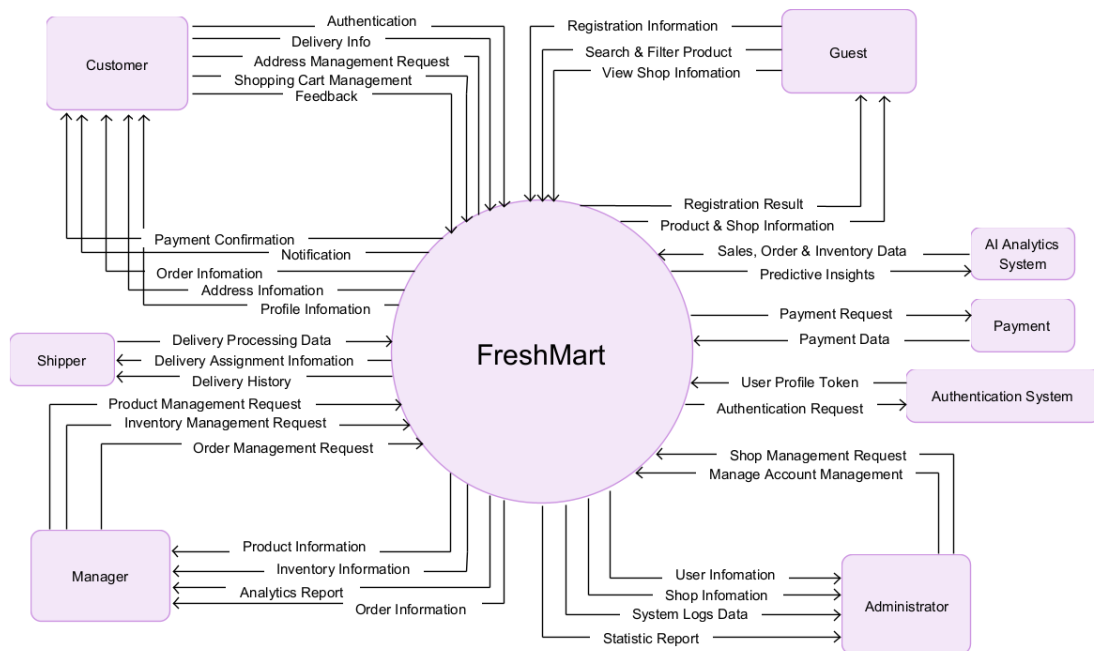
1.1 Context Diagram

The FreshMart System is a web-based multi-vendor online fruit shopping platform designed to support customers in purchasing fruit products within a centralized digital marketplace. The system allows users to browse and search products across multiple shops, view product and shop details, manage shopping carts, place orders, process payments, track deliveries, and submit product reviews or refund requests.

In addition to customer-facing functionalities, **FreshMart** also supports comprehensive business operations for different stakeholder roles. Managers are responsible for managing shop-specific operations, including product management, inventory control, order processing, shipment coordination, and sales monitoring within their assigned shops. Administrators oversee system-wide operations such as shop lifecycle management, user account management, role assignment, category management, and system monitoring. Shippers handle logistics activities including order delivery, shipment status updates, delivery proof submission, and failure reporting.

The system interacts with external services to ensure smooth operation and integration. These include the Authentication System for user authentication and access control, the Payment Gateway for secure online transaction processing, and the AI Analytics System for sales forecasting and business insights. Notification services are also used to deliver real-time updates regarding orders, payments, and system events.

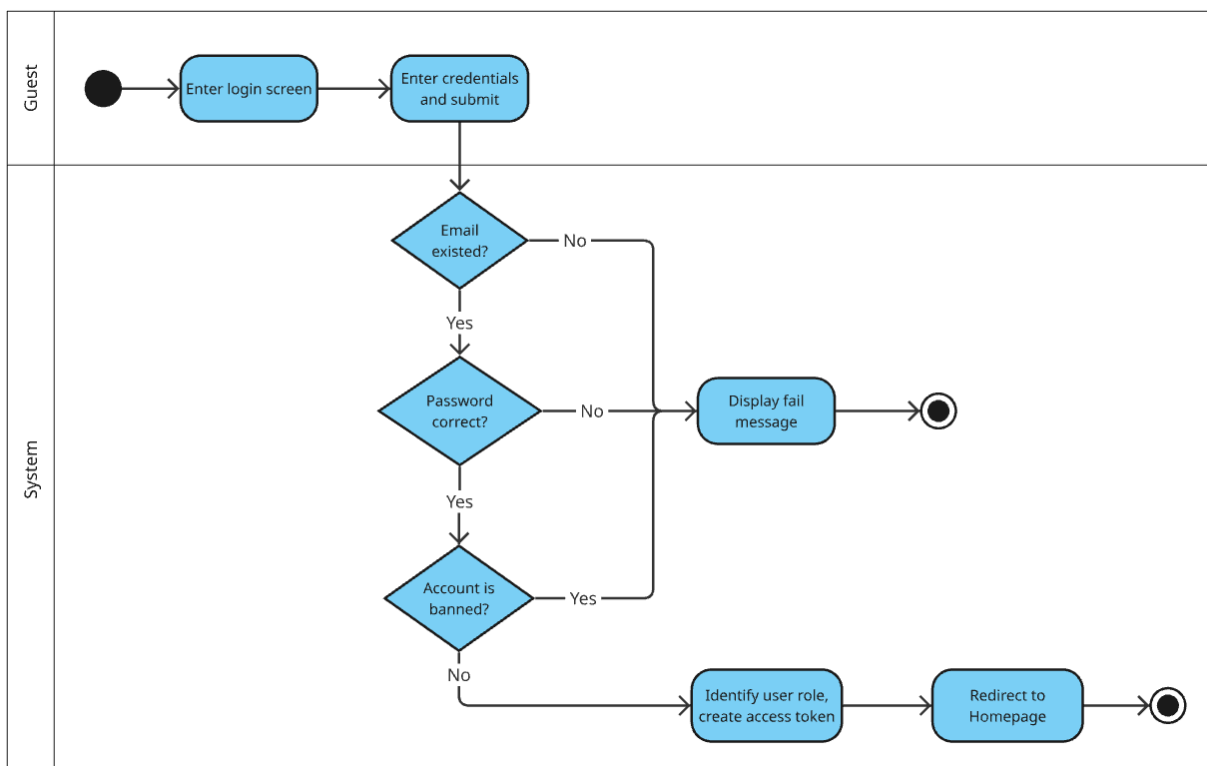
The system is designed as a scalable marketplace platform and is expected to evolve with enhanced automation, improved analytics capabilities, and deeper integration with external payment, logistics, and AI-driven services in future versions.



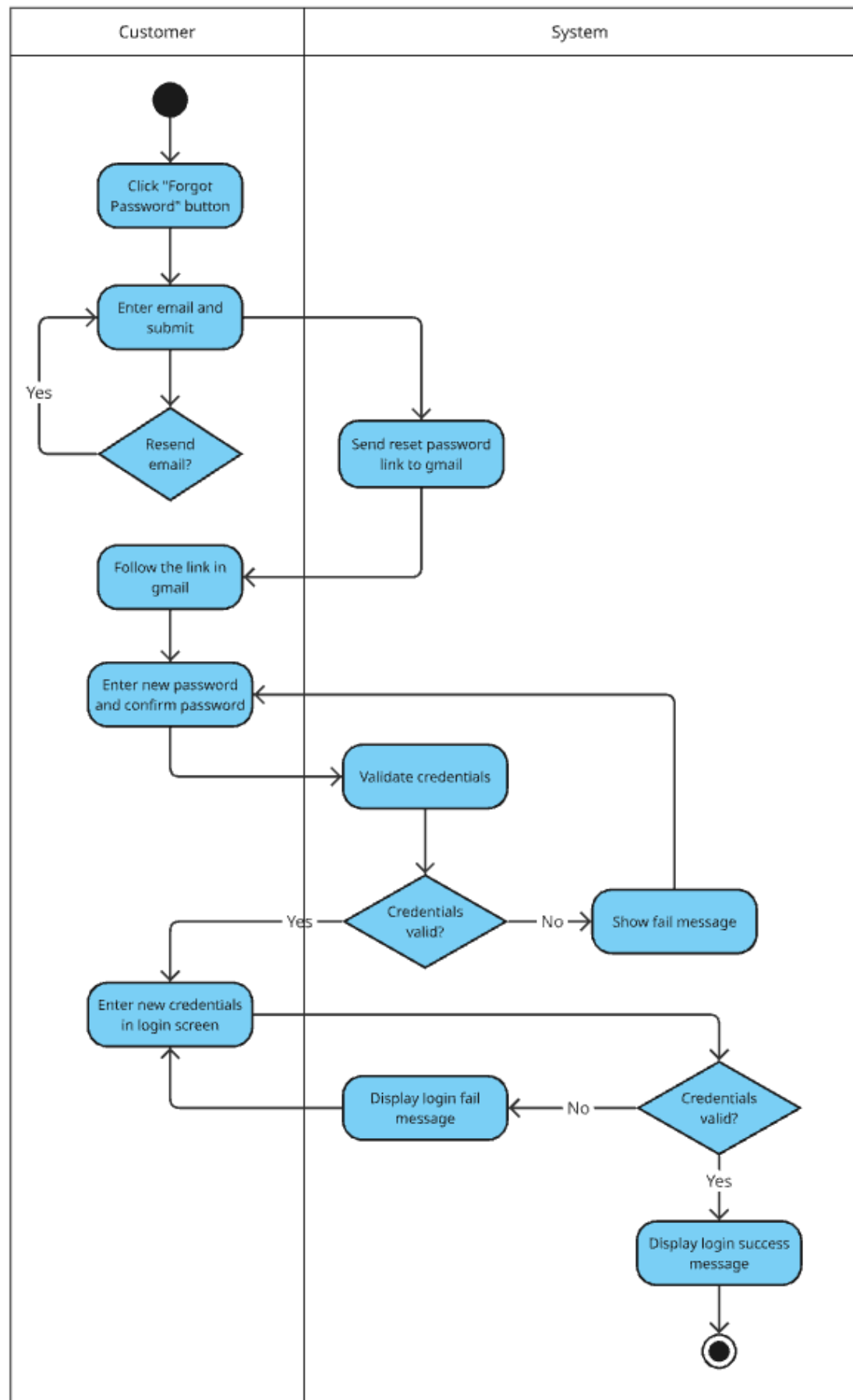
1.2 Main Workflows

1.2.1 Authentication

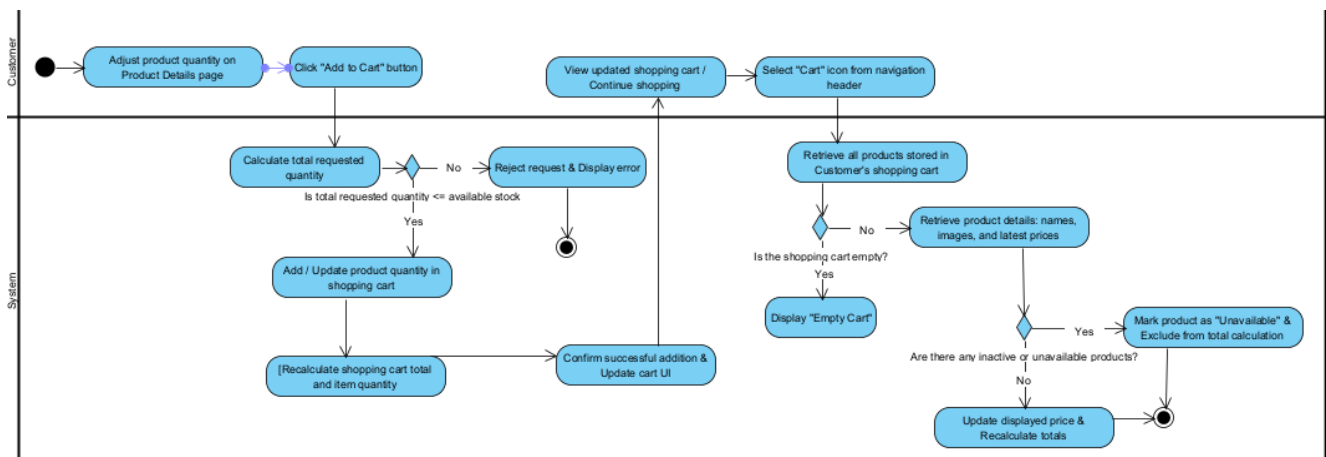
1.2.1.1 Login



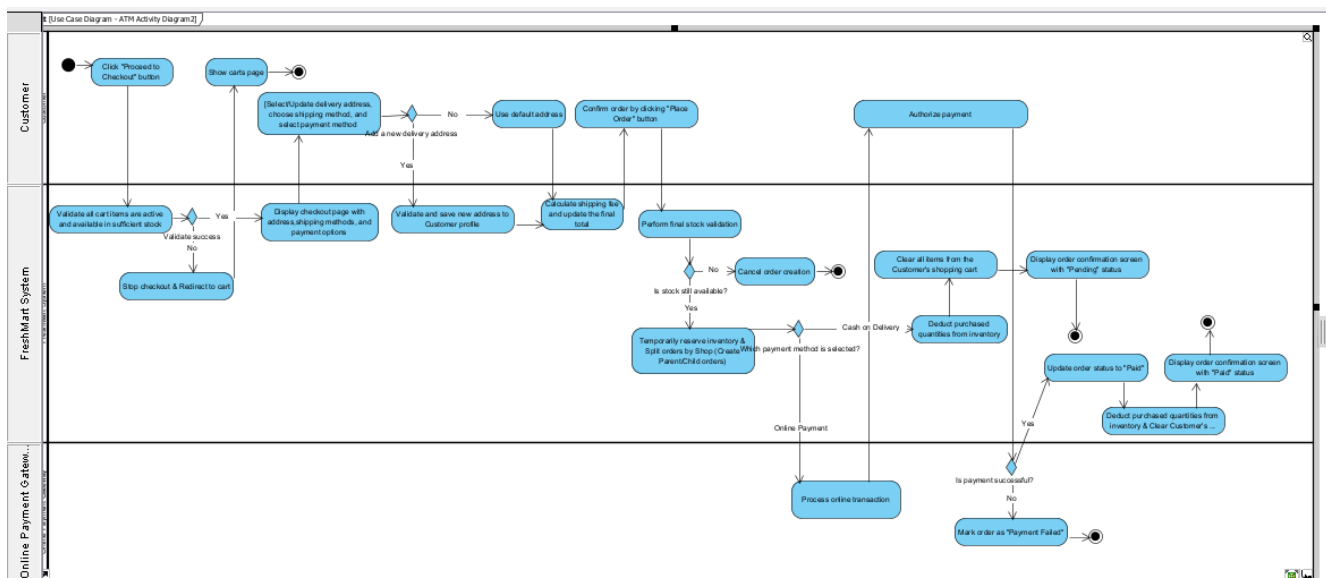
1.2.1.2 Forgot Password



1.2.2 Shopping Cart Management

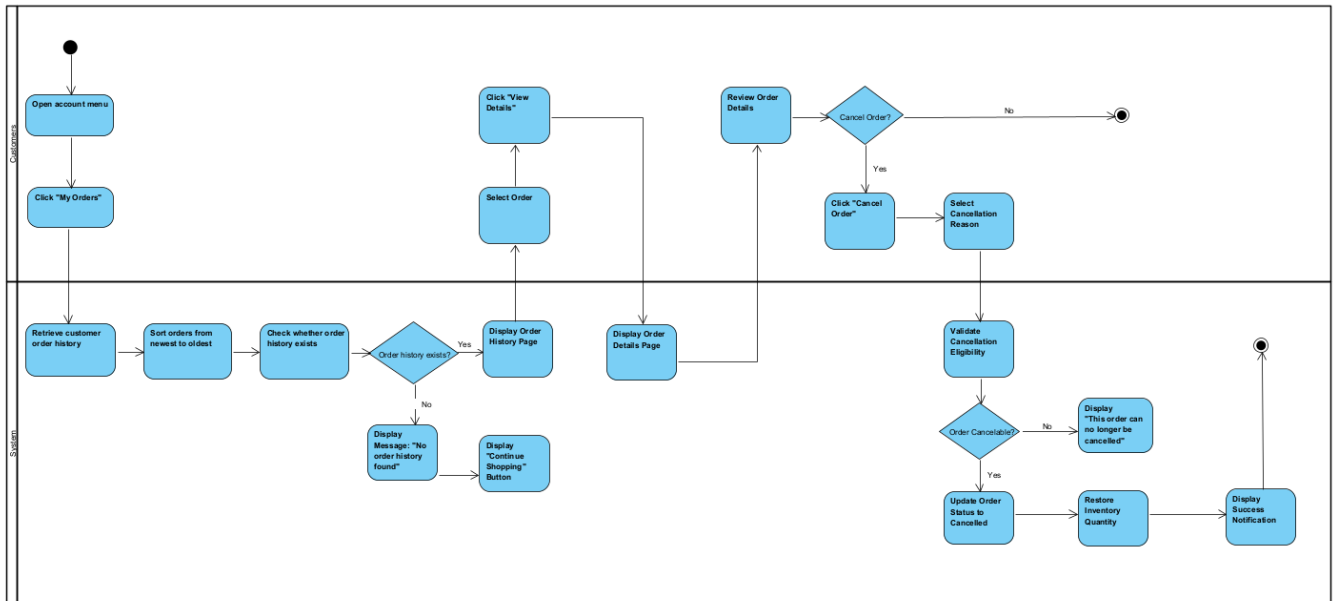


1.2.3 Checkout & Place Order

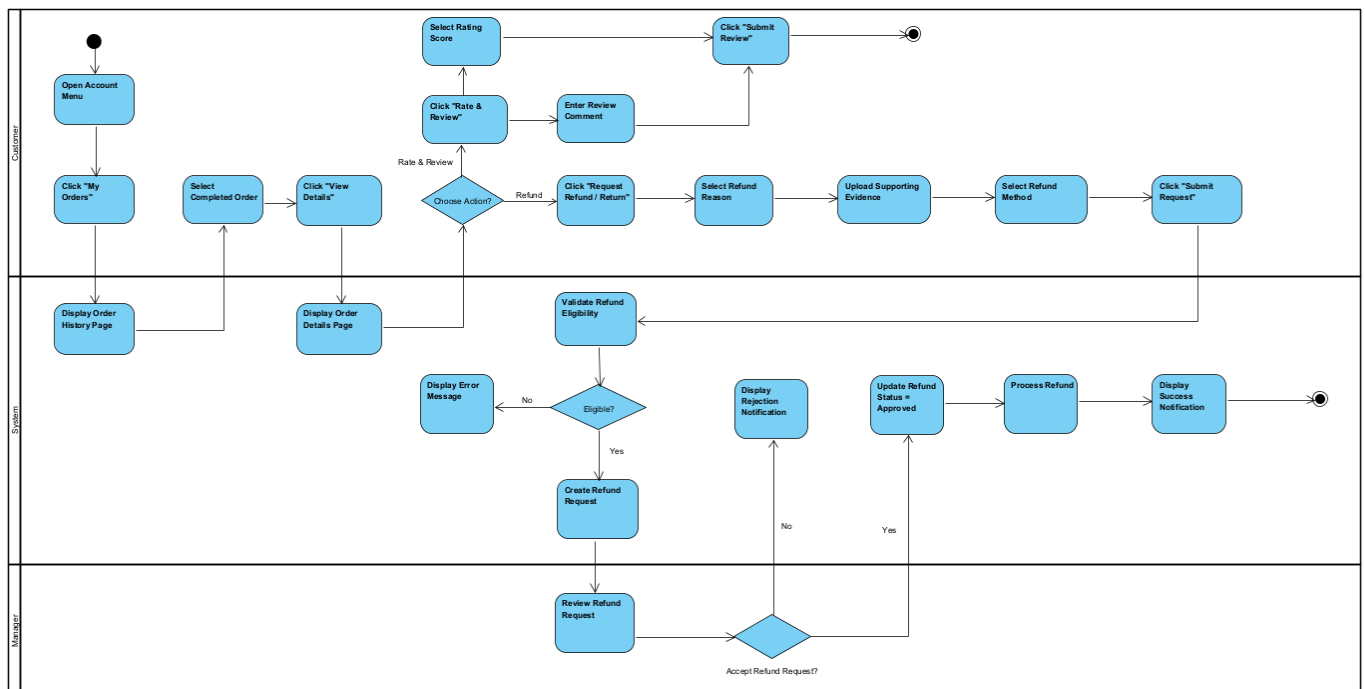


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1.2.4 Order Lifecycle & Tracking

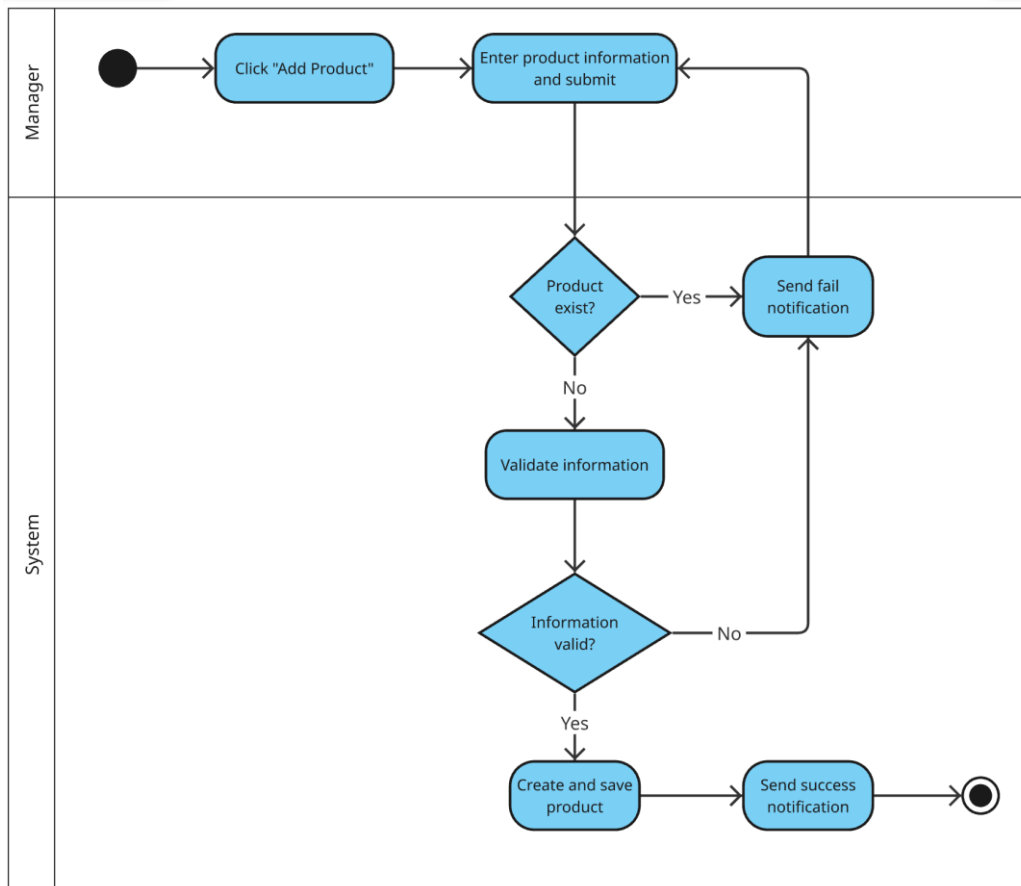


1.2.5 Post-Purchase Services

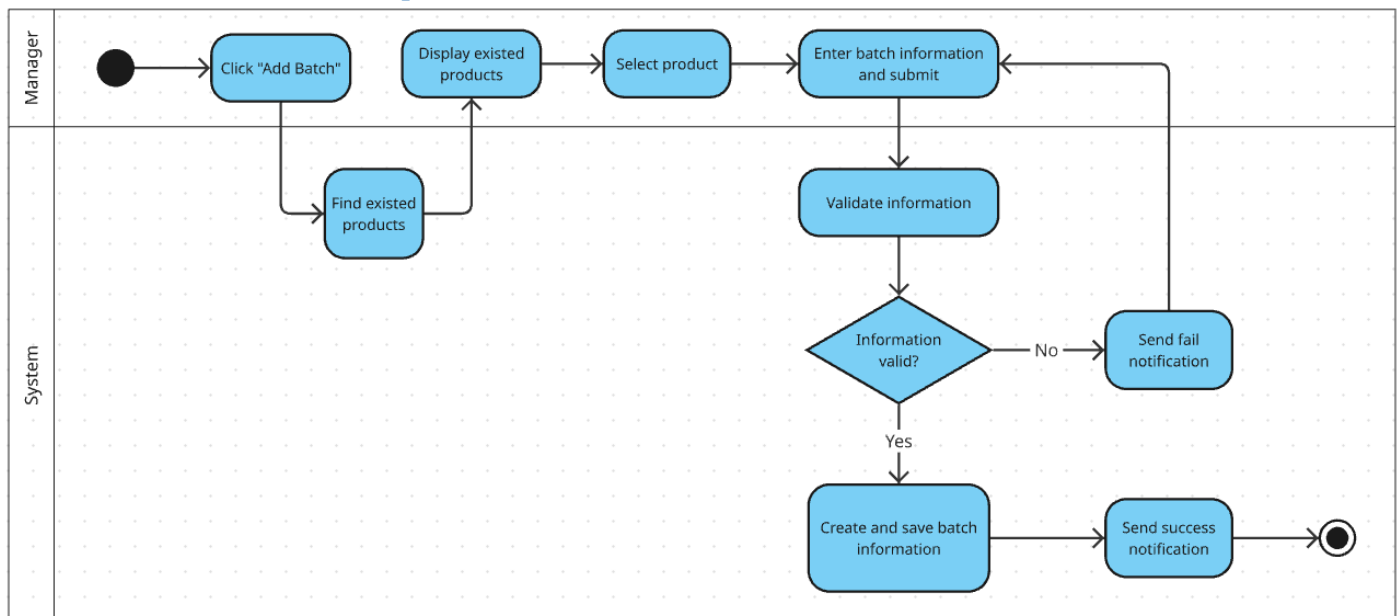


1.2.6 Shop & Product Management

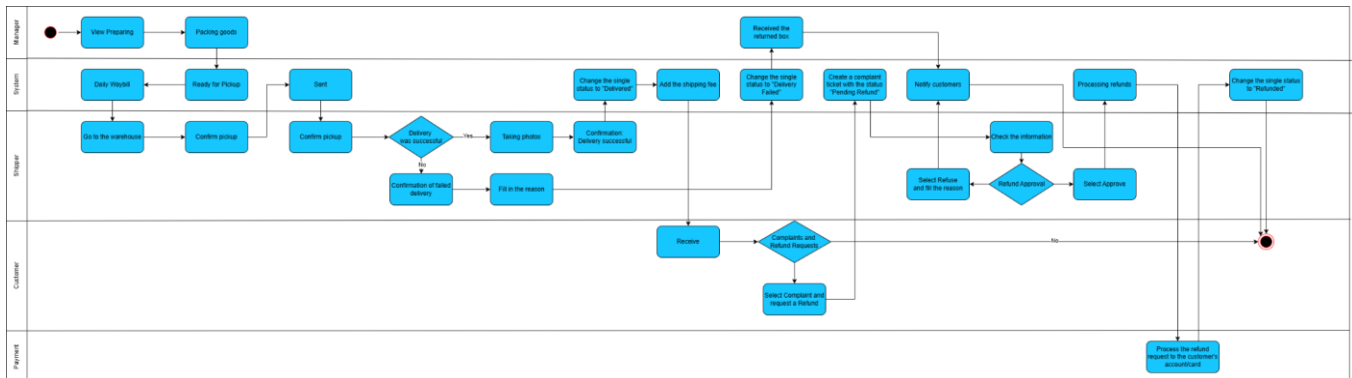
1.2.6.1 Add Product Information



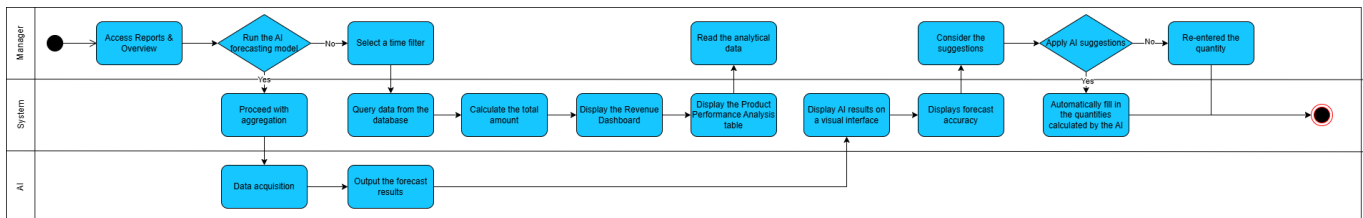
1.2.6.2 Add Batch and Update Product Stock



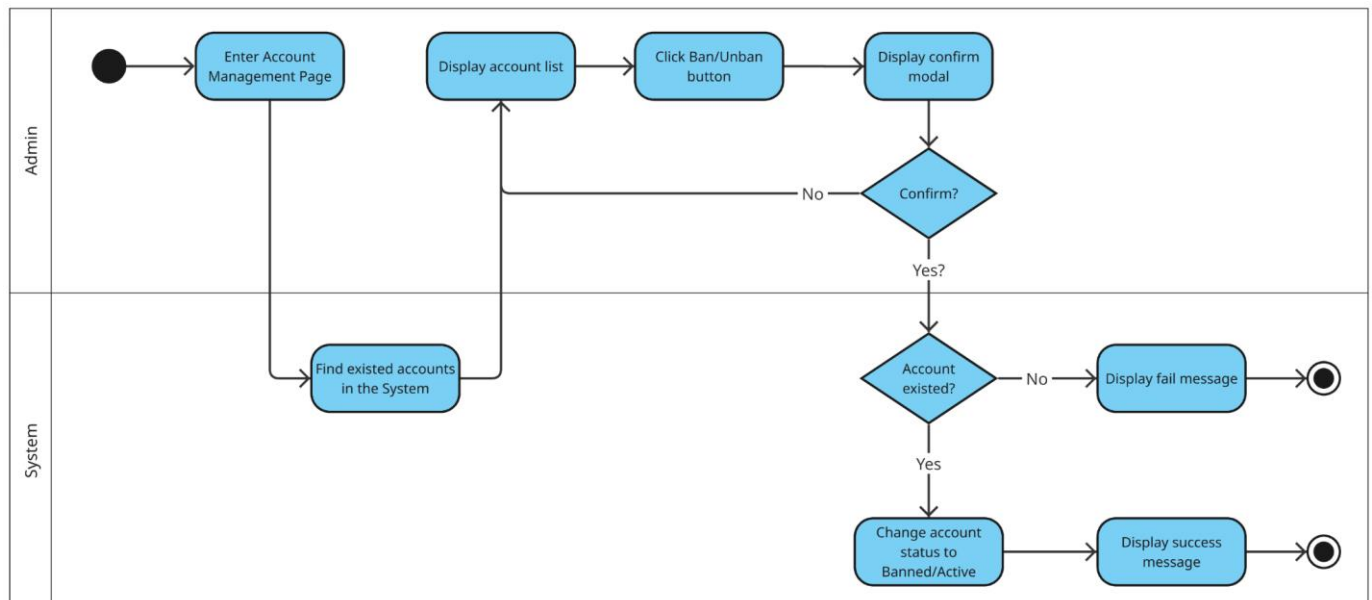
1.2.7 Order Fulfillment & Refund Processing



1.2.8 Analytics & AI Forecasting



1.2.9 Administrator Ban Account



1.3 User Requirements

1.3.1 Actors

#	Actor	Description
1	Guest	An unauthenticated visitor who browses the platform. They can search, filter, and view fruit products, explore shop information, and read system policies before registering or logging in.
2	Manager	A business user responsible for managing one or more assigned shops on the platform. Managers handle product management, inventory control, order processing, refund handling, and sales performance monitoring for their respective shops.
3	Customer	A registered user who purchases products on the platform. Customers can manage shopping carts, add multi delivery addresses, place and pay for orders, track shipment status, and submit reviews or refund requests after purchase.
4	Shipper	A logistics staff member responsible for delivering orders. Shippers can view available delivery tasks, accept orders, update delivery status, upload delivery proof, and report delivery failures.
5	Administrator	System-level user with full administrative privileges. Administrators manage user accounts, roles and permissions, system categories, system logs, and review system-wide reports to ensure compliance and stability.
6	Authentication System	An internal system module responsible for user authentication and session management, including login validation, password recovery, and role-based access control enforcement.
7	AI Analytics System	An internal analytics subsystem that processes historical sales, order, and inventory data to generate predictive insights, including revenue forecasting, demand trends, and stock recommendations for managers across shops.
8	Payment	An external third-party service (e.g., VNPAY, banking APIs, or other payment providers) responsible for processing online transactions and returning payment status results. The system does not store sensitive payment credentials.

1.3.2 Use Case List (UC)

ID	Use Case	Feature	Use Case Description
01	Sign In	Authentication	Allows users (Guest, Customer, Shipper, Manager, Admin) to log into the system using their Email and Password.
02	Sign Up	Authentication	Allows visitors (Guests) to register a new customer account, which saves their credentials into the users table.
03	Forgot Password	Authentication	Enables users to request a password reset link or temporary password via email if they forget it.
04	Change Password	Authentication	Allows authenticated customers to securely change their current password.
05	View and Update Profile	Authentication	Allows customers to view and update their personal information (such as full name, phone number, gender, and date of birth) and manage their account avatar.
06	Manage Customer Accounts	User Management	Allows Admins to view all registered customers and issue account restrictions, such as banning or unbanning accounts violating community terms.
07	Manage Manager Accounts	User Management	Allows Admins to View, Update, Disable, or Delete administrative profiles of corporate Managers.
08	Manage Shipper Accounts	User Management	Allows Admins to evaluate applications, inspect profiles, or Disable/Delete delivery staff accounts .
09	Browse and Search Products	Fruit Browsing	Allows guests/customers to view the product list, search by keywords, or filter products by categories .
10	View Product Details	Fruit Browsing	Displays comprehensive product information including images , description, price, stock availability , and user ratings .
11	Add Product to Cart	Cart Management	Allows customers to add products with a designated quantity into their personal shopping cart.
12	View Shopping Cart	Cart Management	Displays the list of all selected items, their respective quantities, and the current subtotal in the customer's cart.
13	Update Cart Item Quantity	Cart Management	Allows customers to increase or decrease the quantity of specific items directly inside the shopping cart interface.
14	Remove Item from Cart	Cart Management	Allows customers to delete specific products or clear items entirely from their shopping cart.
15	View Order History	Order Tracking	Allows customers to view a comprehensive historical list of all their previously placed orders.
16	View Order Details	Order Details View	Displays specific itemizations, individual pricing, payment status, and purchase timestamps of a single selected historical order.
17	Cancel Order	Order Details View	Allows customers to cancel a placed order, provided its current status is still pending manager confirmation
18	Rate and Review Products	Order Details View	Allows customers to submit a numerical rating and text feedback for products they have successfully purchased .
19	View Managed Shops	Shop Management	Allows managers to view the list of shops assigned to them and select a shop for performing management operations.
20	View Shop Details	Shop Management	Allows administrators and managers to view detailed information of a shop including profile data, status, and business-related information.

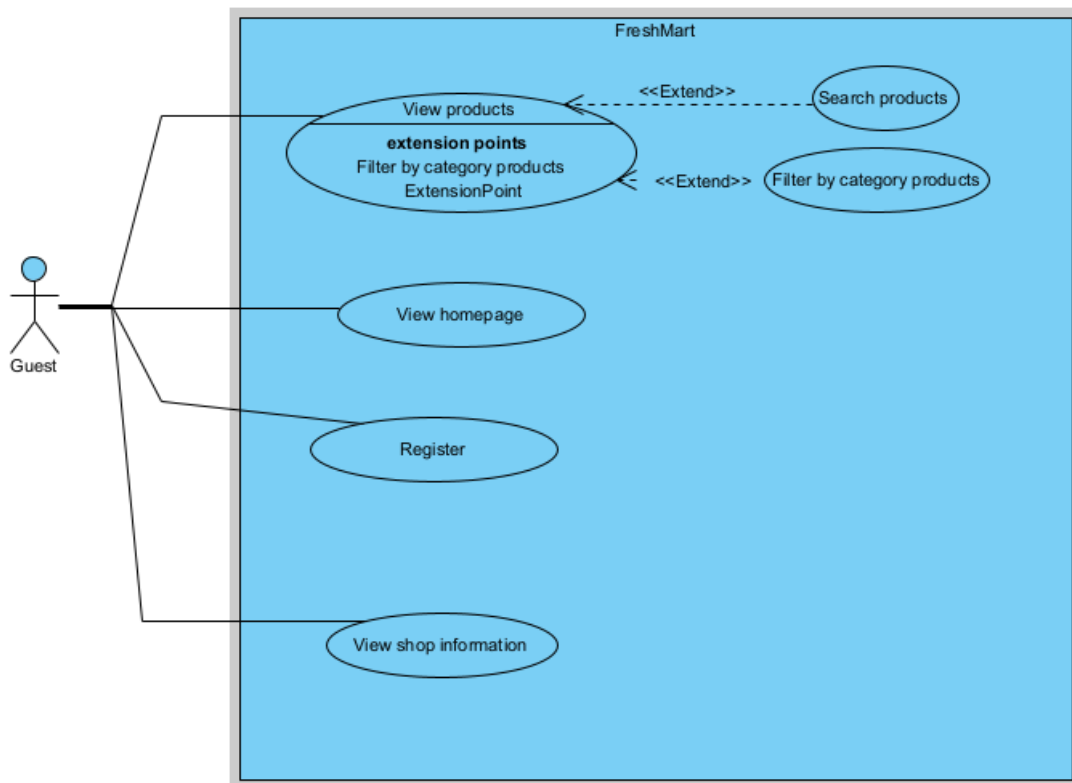
21	View Product Catalog	Category Management	Allows Managers to view and browse the list of all products belonging to their specific shop, showing details like name, price, and current status.
22	Create New Product	Product Management (Manager)	Allows Managers to add a new product to their shop by inputting its name, price, description, and image URL, while selecting (picking) its category to save into product categories.
23	Update Product Catalog	Product Management (Manager)	Allows Managers to edit existing product information (such as changing the price, updating description, or re-assigning categories) to keep the store catalog accurate.
24	Delete Product	Product Management (Manager)	Allows Managers to permanently remove a product from the products table (usually permitted only if the product has no historical orders to maintain data integrity).
25	Toggle Product Availability Status	Product Management	Allows Managers to toggle a product's active visibility to hide seasonal or temporarily out-of-stock items without deleting them.
26	View Batch List	Batch Management	Manager view the list of imported product batches in the system. The batch list helps the Manager monitor stock, price, expiration date, supplier, and batch status.
27	Create Product Batch	Batch Management	Manager create a new imported fruit batch. After creation, the system automatically updates inventory and product price.
28	View Batch Detail	Batch Management	Manager to view detailed information of a specific product batch, including batch data, inventory information, price information, and related history.
29	Update Stock Quantity	Batch Management	Manager manually update the stock quantity of a product batch. Every stock change must be saved in stock history for tracking purposes.
30	Update Selling Price	Batch Management	This use case allows the Manager to update the selling price of a product batch or product. Every price change must be saved in price history
31	View Stock Change History	Batch Management	This use case allows the Manager to view the stock change history of a batch or product.
32	View Price Change History	Batch Management	This use case allows the Manager to view the price change history of a batch or product.
33	Checkout and Place Order	Checkout & Payment	Allows customers to review their cart, select a delivery address, choose a shipping method, select a payment method (Cash or Online), and confirm the order. The system calculates the total cost, updates inventory, and creates records in orders and payments.
34	Request Refund / Return	Order Details View	Allows customers to select a faulty order items, submit a reason, upload proof, choose a return method, and file a claim .
35	View Notifications	Notification Management	Allows customers to view real-time system alerts regarding order updates, account security, and personalized marketing recommendations.
36	View Shop Information	Shop Browsing	Allows users to view detailed business information of a specific vendor/store on the platform.

37	View Website Information	Shop Browsing	Allows visitors to view the platform's introduction, policies, and terms of service.
38	View Address Book	Address Book	Allows customers to view a list of all their saved delivery addresses, highlighting the default address .
39	Add New Address	Update Profile	Allows customers to input and save a new delivery address containing recipient details and geographic location.
40	Update Address Details	Update Profile	Allows customers to modify any existing address information or toggle its status to become the new default shipping location.
41	Delete Address	Update Profile	Allows customers to remove an fault or unwanted address permanently from their profile account.
42	View Delivery History	Shipping Management	Allows Shippers to review their record of successful and failed deliveries aggregated by day, month, or custom periods.
43	View Pending Delivery List	Shipping Management	Allows Shippers to browse a list of unassigned or pending orders within their assigned operational sectors.
44	Accept Delivery Order	Shipping Management	Allows Shippers to select and self-assign a specific order to take responsibility for its physical delivery.
45	View Delivery Specifications	Shipping Management	Provides Shippers with recipient details: name, phone number, physical address, cash collection amount (COD), tracking code, and special delivery notes.
46	Update Order Delivery Status	Delivery Execution	Allows Shippers to update delivery milestones: "Not Picked Up", "Picked Up & Out for Delivery", "Delivered", or "Delivery Failed".
47	Upload Delivery Proof	Delivery Execution	Allows Shippers to capture and upload a photographic confirmation confirming successful package drop-off.
48	Record Delivery Failure Reason	Delivery Execution	Allows Shippers to log precise reasons for an unsuccessful delivery (e.g., recipient unreachable, incorrect address) and state the change reason.
49	Process Pending Orders	Order Management	Allows Managers to view newly placed customer orders and choose to either Confirm (Accept) or Reject them.
50	Manage Daily Waybills / Shipments	Order Management	Allows Managers to track and organize batches of confirmed orders that have been successfully handed off to third-party logistics or internal Shippers.
51	Inspect Order Processing Details	Order Management	Displays customer profiles, item lists, ordered quantities, total costs, and payment confirmation statuses so the manager can prepare shipments.
52	Update Internal Order Progress	Order Management	Allows Managers to advance the internal fulfillment cycle status, such as changing orders from "Preparing" (packaging) to "Sent" (shipped).
53	Moderate Refund Claims	Order Management	Allows Managers to evaluate customer refund statements and uploaded media proof to officially Approve or Deny the transaction reimbursement .
54	View Revenue Overview Dashboard	Revenue Analytics	Displays gross revenue metrics, total sales volume, and business performance analytics across selected intervals (daily, monthly, annually).
55	Analyze Revenue by Product / Category	Revenue Analytics	Generates statistical breakdowns and visual charts tracking revenue generated by specific merchandise or categories.

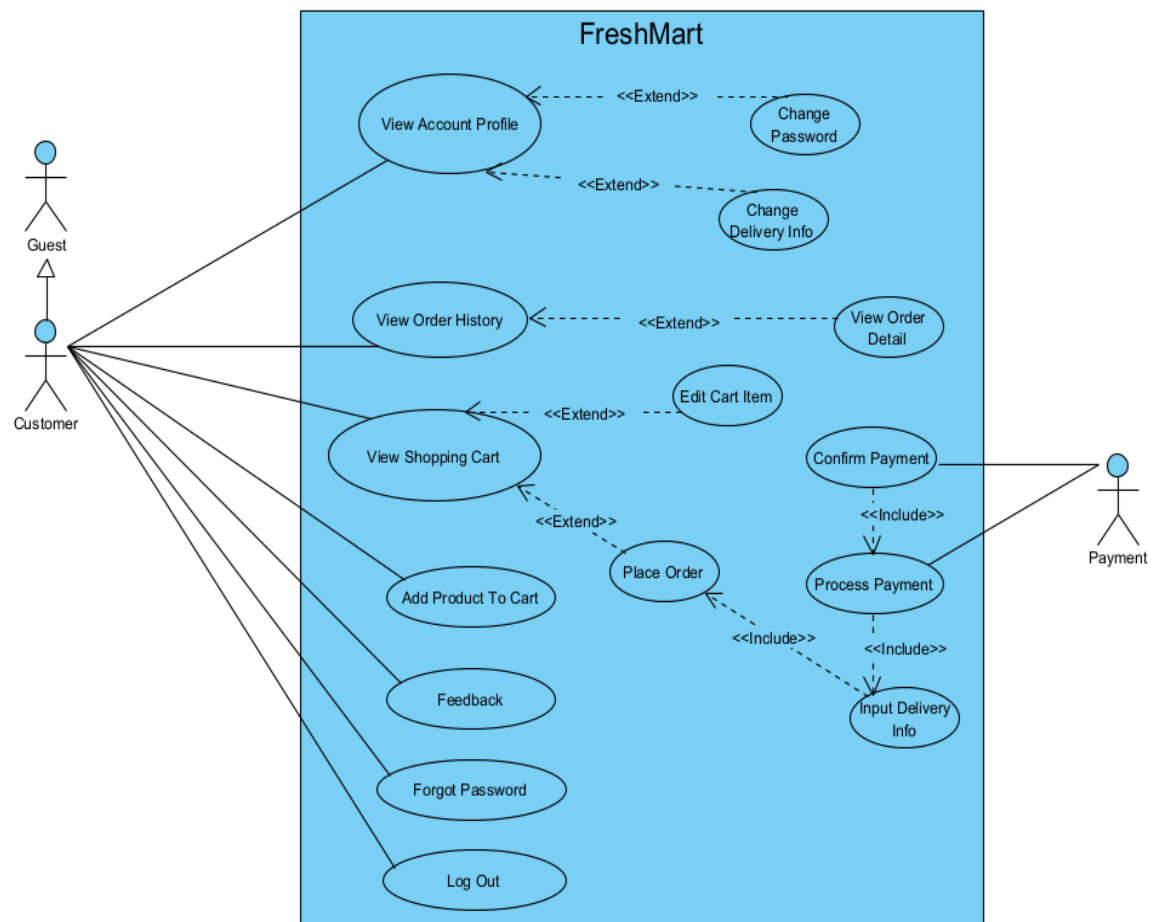
56	Generate AI Sales Forecasts	AI Analytics	Utilizes predictive AI processing to evaluate historical sales logs and project revenue trends and inventory demands for the upcoming month.
57	Receive Monthly User Activity Reports	Analytics & Reporting	Automated system functionality that compiles registration statistics and emails an monthly summary report of new user registrations to the Admin.
58	Inspect System Logs	System Administration	Allows Admins to view system logs to evaluate backend performance, diagnose server traffic bottlenecks, check errors, or audit malicious activity by ip address.
59	View System Categories	Category Management	Allows Admins to view and inspect the entire system-wide product category tree.
60	Create System Category	Category Management	Allows Admins to add a new standardized category for all shops to use.
61	Update System Category	Category Management	Allows Admins to edit the name or description of an existing global category to maintain marketplace organization.
62	Delete System Category	Category Management	Allows Admins to remove a category from the system if it contains no active associated products.
63	Create Shop	Shop Management	Allows administrators to create a new shop in the system with required information such as shop name, description, address, contact details, and operational status.
64	View Shops List	Shop Management	Allows administrators or managers to view a list of all shops in the system or only assigned shops, including basic shop information and status.
65	Update Shop	Shop Management	Allows administrators or authorized managers to update existing shop information such as name, description, address, contact details, and status.
66	Deactivate/Delete Shop	Shop Management	Allows administrators to deactivate or remove a shop from the system if it has no active products or related orders.
67	Manage Shop	Shop Management	Allows managers to manage shop-level operations including product management, order processing, and inventory control within their assigned shops.
68	Assign Shop to Manager	Shop Assignment	Allows administrators to assign one or multiple shops to a manager for operational responsibilities such as product, order, and inventory management.

1.3.3 Use Case Diagrams

1.3.3.1 UCs for Guest



1.3.3.2 UCs for Customers



1.3.3.3 UCs for Manager

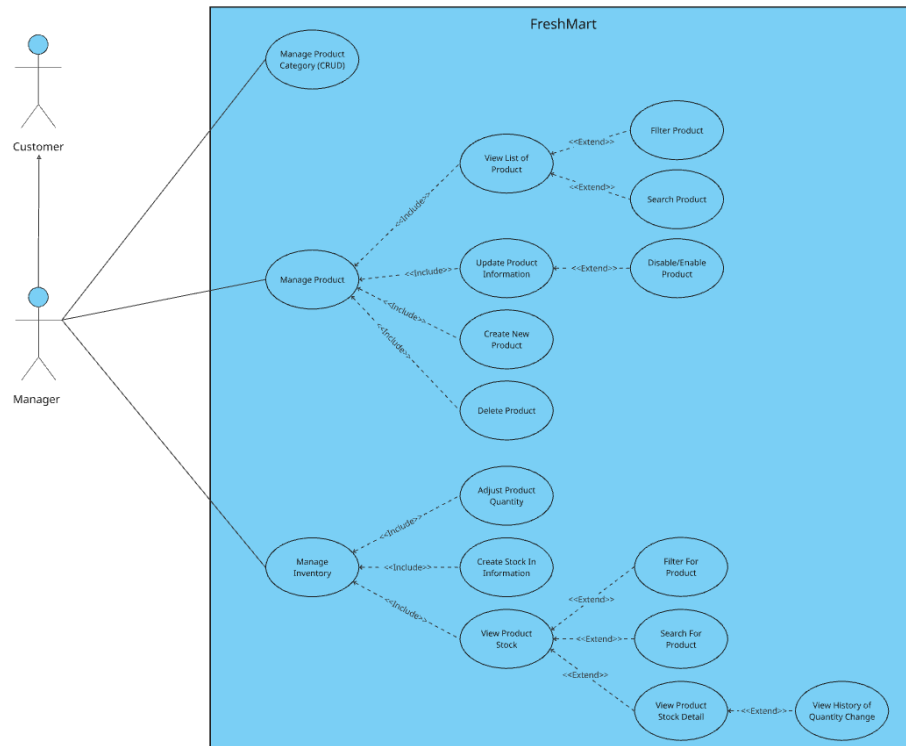


Figure 1.3.3.3.1 UC for Manager

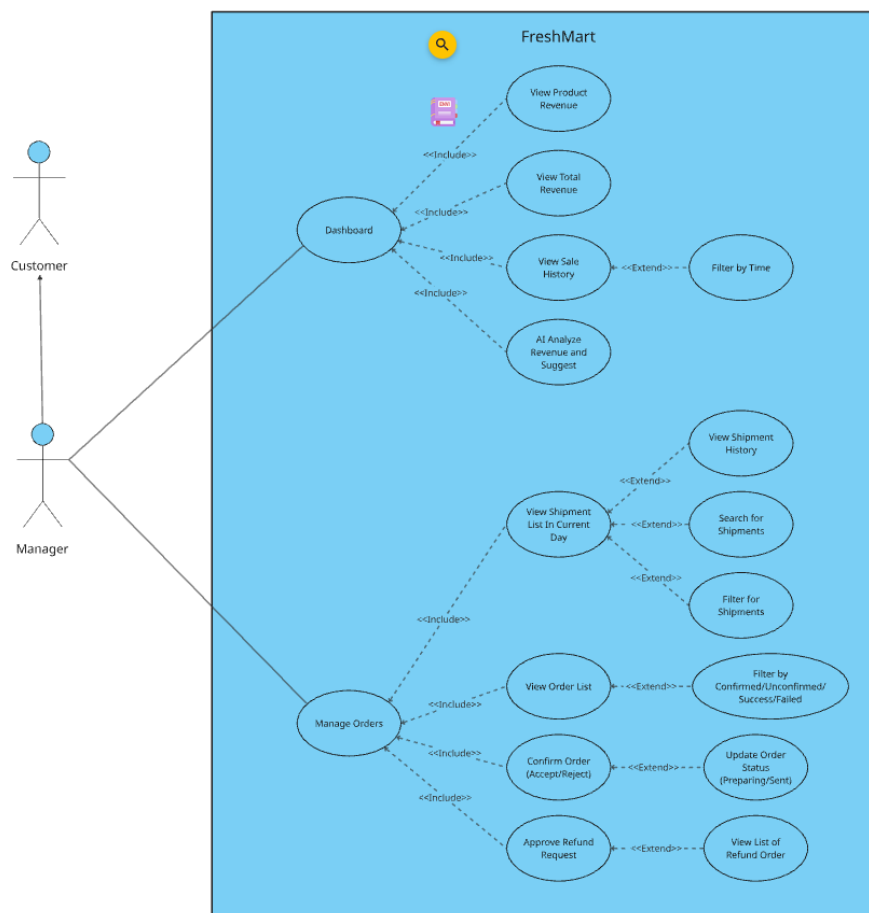
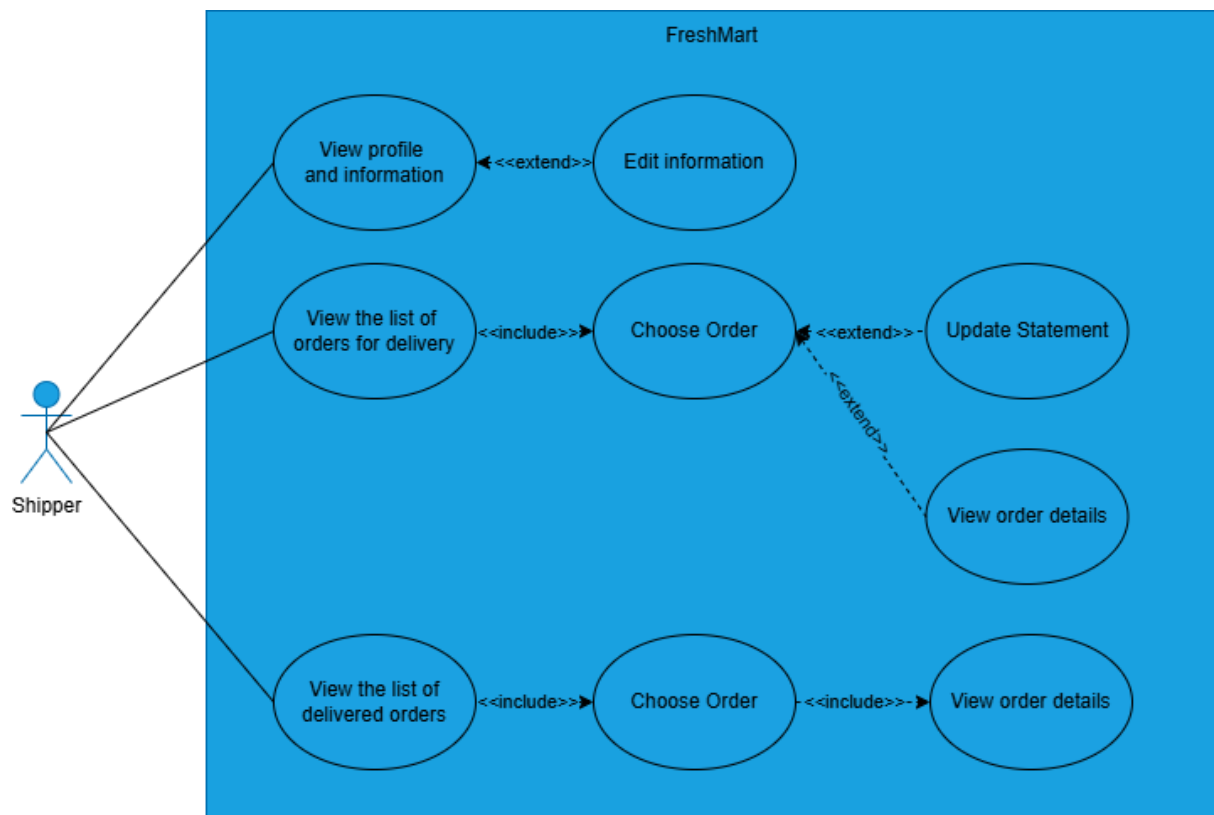
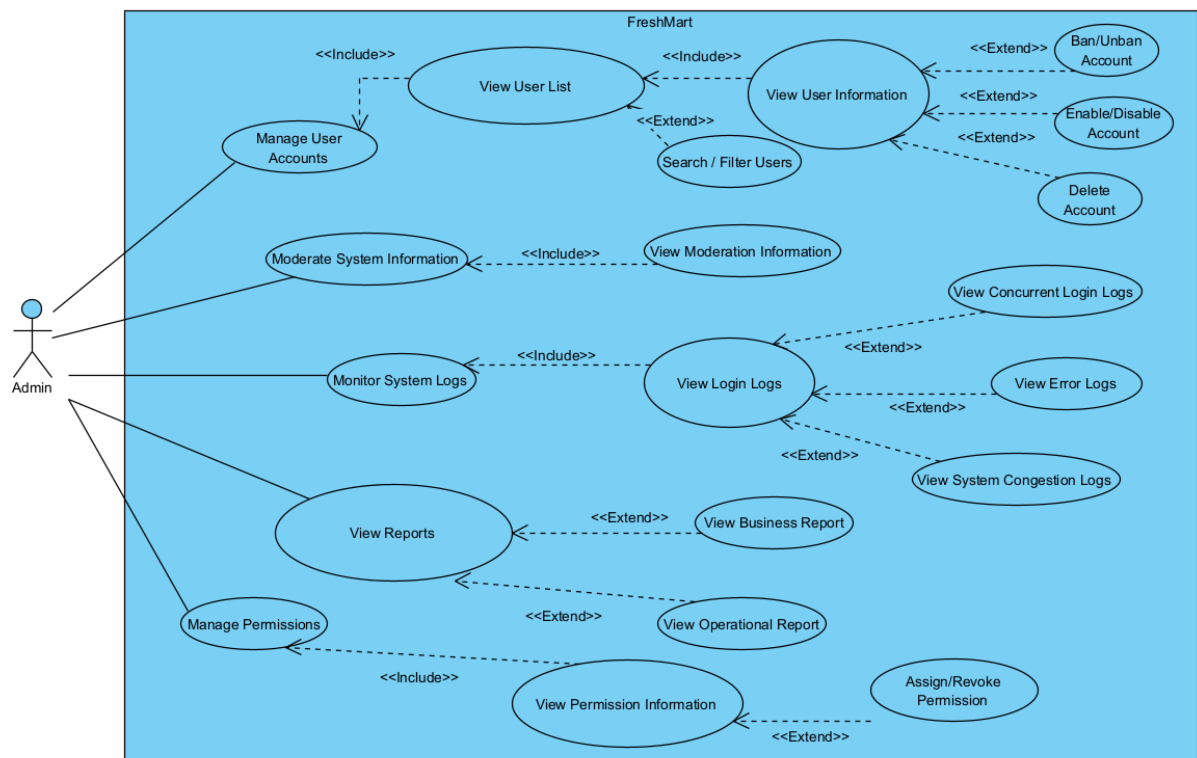


Figure 1.3.3.3.2 UCs for Manager

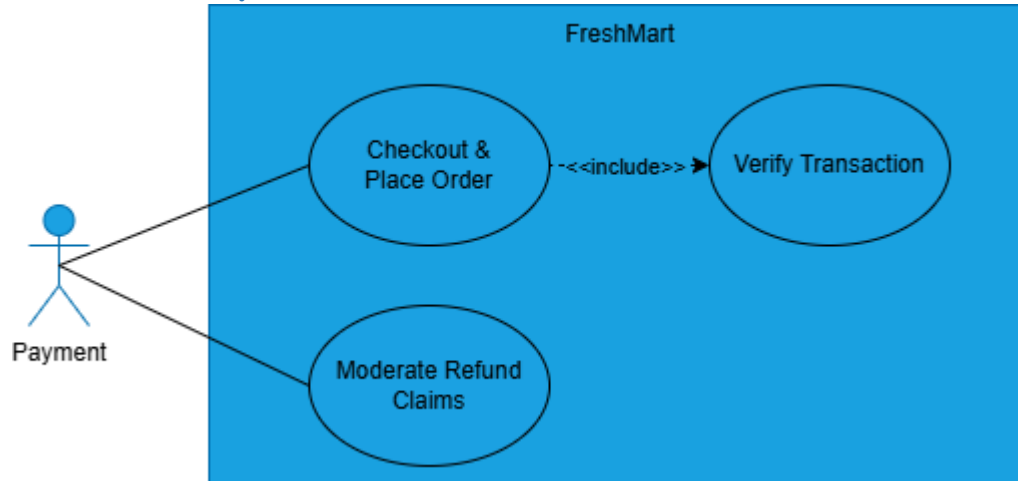
1.3.3.4 UCs for Shipper



1.3.3.5 UCs for Admin



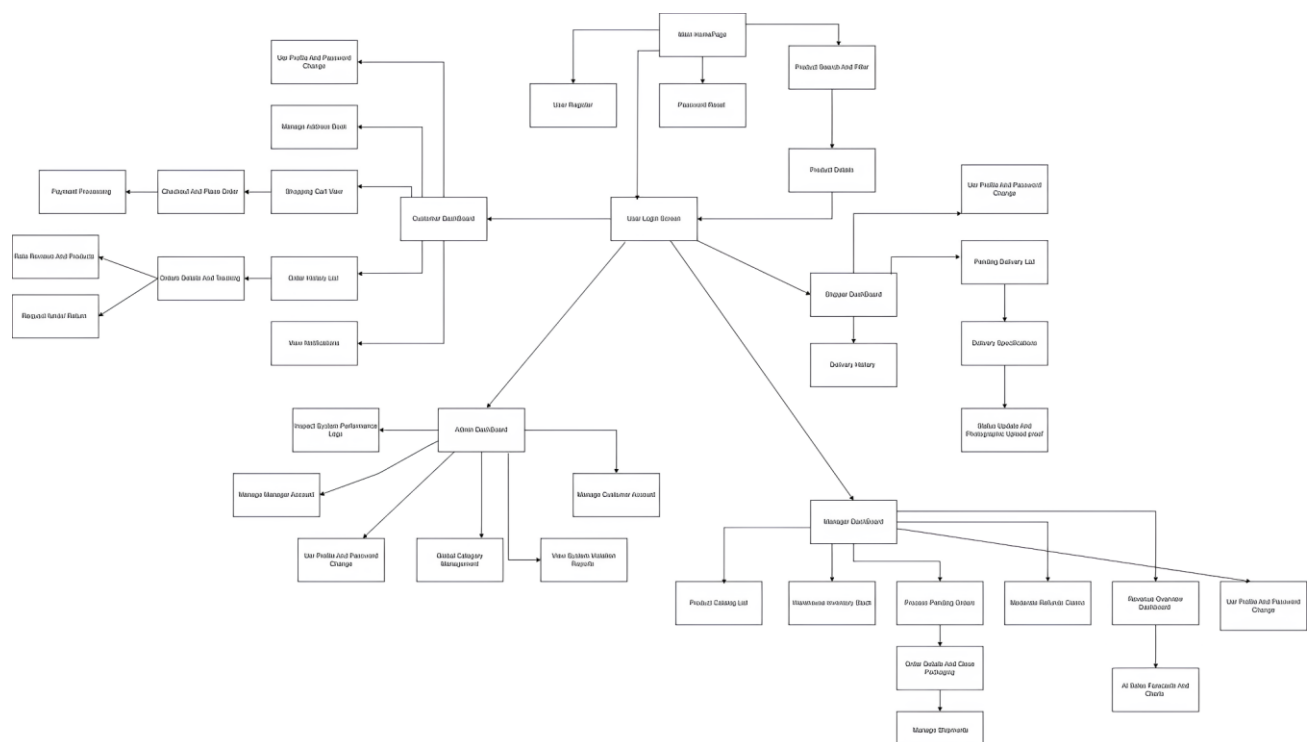
1.3.3.6 UCs for Payment



1.4 System Functionalities

1.4.1 Screens Flow

1.4.1.1 Overall screenflow



[View Detail](#)

1.4.2 Screen Authorization

Screen	Admin	Guest	Customer	Shipper	Manager
Main HomePage	X	X	X	X	X
Login / Register / Forgot Password	X	X	X	X	X
User Profile And Password Change	X	X	X	X	X
Product Search And Filter		X	X		
Product Details		X	X		
Customer Dashboard			X		
Manage Address Book			X		
Shopping Cart View			X		
Order History List			X	X	X
View Notifications	X	X	X	X	X
View Order Details And Tracking			X	X	
Rate Reviews And Products			X		
Request Refund Screen			X		
Checkout And Place Order			X		
Payment Process			X		
Admin DashBoard	X				
System Performance Logs	X				
Manage Manager Account	X				
Global Category Management Screen	X				
System Violation Report	X				
Manage Customer Account	X				
Manager DashBoard					X
Product Catalog List Screen					X
Warehouse Inventory Stock Screen					X
Pending Orders Processing Screen					X

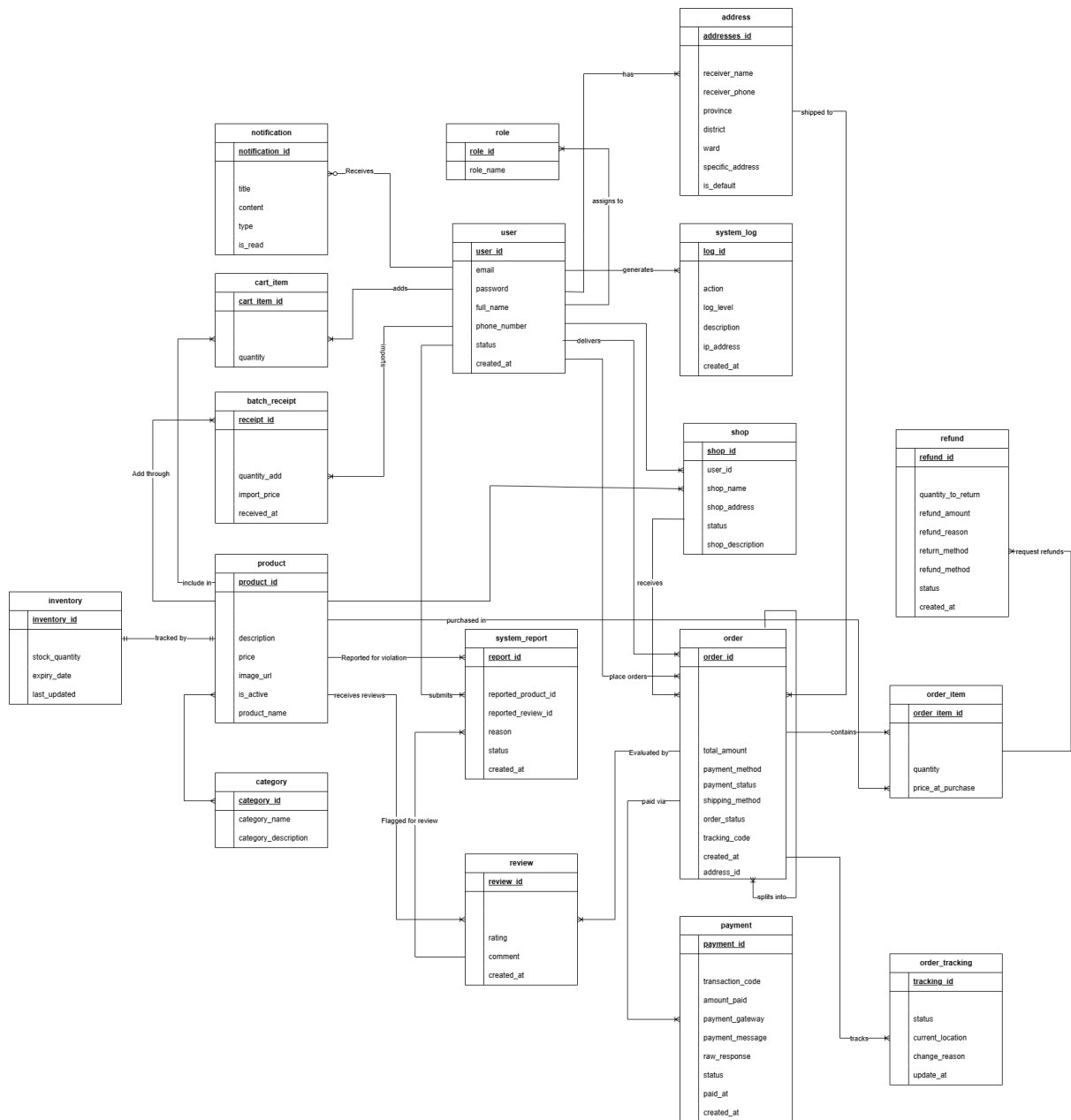
Order Details And Close Packaging					X
Manage Shipments					X
Moderate Refund claims Screen					X
Revenue Overview Dashboard					X
AI Sales Forecasts And Charts					X
Shipper DashBoard				X	
Delivery History Screen				X	
Pending Delivery List Screen				X	
Pending Delivery Specifications Screen			X	X	
Status Update And Photographic Upload proof Screen				X	

1.4.3 Non-UI Functions

#	Feature	System Function	Description
1	Authentication & Account	Password Recovery Service	Sends password reset emails/OTP verification to Customers during password recovery requests.
2	Notification Management	Real-time Notification Dispatcher	Automatically pushes notifications to Customers when order status, refund status, or account-related events change.
3	Order Management	Inventory Synchronization Service	Automatically deducts or restores fruit inventory quantities when orders are placed, cancelled, or refunded.
4	Checkout & Payment	Payment Gateway Integration Service	Communicates with external payment providers (e.g., VNPAY, PayOS) to process online payment requests and receive payment status callbacks.
5	Checkout & Payment	Payment Status Synchronization Job	Periodically synchronizes pending online payment statuses from external payment providers into the system database.
6	Refund Management	Refund Validation Service	Validates refund eligibility based on refund period, order completion status, and duplicate request checks.
7	Dashboard & Report	Revenue Aggregation Service	Periodically aggregates sales transactions to generate dashboard metrics such as total revenue, order counts, and growth trends.
8	AI Forecasting	AI Sales Forecast Engine	Analyzes historical sales and inventory data to generate predictive revenue and stock recommendations for managers.

9	Reporting	Monthly User Activity Report Scheduler	Automatically executes a scheduled monthly task to compile user registration statistics and email reports to Administrators.
10	System Monitoring	System Log Collection Service	Continuously records system events, login activities, security incidents, and application errors into centralized logs.
11	System Monitoring	Security Alert Detection Service	Detects abnormal activities (e.g., repeated failed login attempts, suspicious IP requests) and flags them for administrator review.
12	Product Management	Expiration Monitoring Service	Periodically checks fruit expiration dates and flags products approaching expiration or automatically disables expired products.
13	Category Management	Category Hierarchy Validation Service	Validates parent-child category relationships to prevent circular dependencies or invalid hierarchy assignments.
14	Notification Management	Notification Read-State Synchronizer	Updates notification read/unread status and synchronizes unread badge counters across sessions.
15	Order Tracking	Order Status Update Service	Automatically updates order states based on manager/shipper actions and triggers corresponding notifications.

1.5 Entity Relationship Diagram



[Link diagram](#)

Entities Description

#	Entity	Description
1	Roles	Stores system authorization roles including Admin, Manager, Shipper, and Customer.
2	Users	Contains master account credentials, personal profiles, and account statuses.
3	UserRoles	Associative table establishing the many-to-many relationship between users and roles.
4	Shops	Manages individual storefront profiles owned and operated by store managers.

5	Addresses	Keeps track of multiple shipping addresses saved by customers for order deliveries.
6	Categories	Defines product classifications and groupings used across the fruit platform.
7	Products	Contains detailed commercial data for individual items listed for sale by shops.
8	ProductCategories	Associative table establishing the many-to-many relationship between products and categories.
9	Inventory	Tracks active stock quantities, expiration dates, and updates for products (1:1 relation).
10	InventoryReceipts	Tracks product import history and original purchase costs.
11	CartItems	Holds temporary lists of products and intended quantities selected by shoppers before checkout.
12	Orders	Logs transaction headers including split order structures, total billing, and fulfillment states.
13	OrderItems	Stores the final list, quantity, and locked price of products for each order
14	OrderTracking	Provides a real-time chronological log of delivery milestones and geographic updates.
15	Payments	Serves as the immutable financial ledger logging bank transaction codes and settlement states.
16	Reviews	Stores user-generated ratings and comments verifying post-purchase product feedback.
17	Refunds	Handles transactional item returns, claim evaluations, and repayment tracking details.
18	SystemLogs	Maintains an audit trail of user activities, administrative actions, and system error events.

3. Use Case Specifications

3.1 Authentication & Access Management

- 3.1.1 UC 01: Sign In

ID and Name	UC01 - Sign In		
Created by	Trình Minh Quang		
Primary Actor	Guest	Secondary Actor	None
Description	Allows users (Guest, Customer, Shipper, Manager, Admin) to log into the system using their registered Email and Password to access authorized roles.		
Preconditions	1. The user has an active account in the system database.		
Main Sequence	1. The guest selects the "Sign In" option from the navigation bar. 2. The system displays the login form requesting Email and Password. 3. Guests enter their registered Email and Password. 4. Guest clicks the "Login" button to submit credentials. 5. The system validates the email format and checks the credentials against the users table. 6. The system identifies the user's role and initiates the session. 7. The system confirms successful authentication and redirects the user to their respective dashboard or homepage.		
Alternative Sequences	Step 5: If the Email or Password field is left empty, the system displays a validation error message. Step 5: If the credentials do not match any record in the database, the system rejects the request and displays: "Incorrect email or password." Step 6: If the account status is flagged as "Banned" or "Disabled", the system terminates the login process and displays a notification to contact support.		
Post conditions	The user is successfully authenticated, and the system session is established based on the user's role.		
Business Rule	BR-01, BR-02, BR-04, BR-05		

• 3.1.2 UC 02: Sign Up

ID and Name	UC02 - Sign Up		
Created by	Trình Minh Quang		
Primary Actor	Guest	Secondary Actor	None
Description	Allows a visitor (Guest) to register a new account on the platform, saving their credentials into the database as a Customer.		
Preconditions	1 The Guest has accessed the Sign Up page.		
Main Sequence	1. Guest selects the "Sign Up" option from the interface. 2. The system displays the registration form (Full Name, Email, Password, Confirm Password, Phone Number). 3. Guest inputs the required registration information. 4. The guest clicks the "Register" button to submit the form. 5. The system validates the input data format (email pattern, password security strength). 6. System checks if the entered Email already exists in the database. 7. The system creates a new user record with the role set to "Customer" and sets the account status to "Active". 8. The system displays a success notification and automatically logs the user in or redirects them to the Sign In page.		
Alternative Sequences	Step 5:If any mandatory field is missing, the system prompts the Guest to complete all required inputs. Step 5:If the Password and Confirm Password fields do not match, the system displays an error message and requests a correction. Step 6:If the Email is already registered in the system, the system rejects the request and displays: "Email already exists. .		
Post conditions	A new Customer account is successfully created in the system database.		
Business Rule	BR-02, BR-04, BR-05		

• **3.1.3 UC 03: Forgot Password**

ID and Name	UC03 - Forgot Password		
Created by	Trình Minh Quang		
Primary Actor	Customer	Secondary Actor	None
Description	Enables registered users to request a password reset link or temporary password via email if they forget their credentials.		
Preconditions	1. The user has an active registered account in the system. 2. The user has accessed the "Forgot Password" page from the login interface.		
Main Sequence	1. The customer selects the "Forgot Password" option. 2. The system displays the password recovery form requesting the registered Email. 3. The customer enters their registered Email address and clicks "Submit". 4. The system validates the email format and verifies its existence in the database. 5. The system generates a temporary password or a secure reset token link. 6. The system triggers an automated email to the user and displays a success notification.		
Alternative Sequences	Step 4: If the input field is empty or the email format is invalid, the system displays an error message. Step 4: If the email does not exist in the database, the system rejects the request and displays: "Email address not found."		
Post conditions	A secure password reset link or temporary password is successfully generated and dispatched to the user's registered email.		
Business Rule	BR-04, BR-05		

• 3.1.4 UC 04: Change Password

ID and Name	UC4 - Change Password		
Created by	Đỗ Nguyễn Hùng		
Primary Actor	Customer	Secondary Actor	None
Description	Allows a logged-in Customer to update their account password to improve account security.		
Preconditions	2. The Customer is logged into the system. 3. The Customer account is active. 4. The Customer has accessed the Change Password page from Account Settings.		
Main Sequence	9. Customer selects the “Change Password” option from Account Settings. 10. System displays the password update form. 11. Customer enters the current password, new password, and password confirmation. 12. Customer submits the password update request by clicking the “Save Changes” button. 13. System validates the current password and new password format. 14. System verifies whether the new password and confirmation password match. 15. System updates the Customer account password. 16. System confirms successful password update and displays a success notification.		
Alternative Sequences	Step 5: If the entered current password is incorrect, the system rejects the request and prompts the Customer to re-enter the correct password. Step 5: If the new password does not satisfy password requirements, the system rejects the request and displays password requirement instructions. Step 6: If the new password and confirmation password do not match, the system displays a validation message and requests correction. Step 4: Customer may cancel the password update request. The system terminates the process without changing the current password.		
Post conditions	The Customer account password is updated successfully and applied to future login sessions.		
Business Rule	BR-04, BR-05, BR-06		

• **3.1.5 UC 05: View and Update Profile**

ID and Name	UC05 - View and Update Profile		
Created by	Đỗ Nguyễn Hùng		
Primary Actor	Customer	Secondary Actor	None
Description	Allows a logged-in Customer to view and update personal profile information, including full name, email, phone number, profile avatar, and preferred default delivery address.		
Preconditions	1. The Customer is logged into the system. 2. The Customer account is active and accessible. 3. The Customer is currently on any page containing the user profile menu.		
Main Sequence	1. The Customer clicks on the "My Profile" option from the account menu. 2. The system retrieves the Customer's current profile information from the database. 3. The system redirects the Customer to the Profile page and displays existing personal information. 4. The Customer modifies profile information (e.g., full name, phone number, email, avatar, default address). 5. The Customer clicks the "Save Changes" button. 6. The system validates the entered information format and completeness. 7. The system updates the Customer profile information. 8. The system displays a success notification and refreshes the profile page with updated information.		
Alternative Sequences	Step 6: If entered profile information is invalid (e.g., invalid email format or missing required fields), the system highlights invalid fields and requests correction. Step 6: If the entered phone number is already associated with another account, the system rejects the update and displays the message: "Phone number already exists." Step 5: Customers may cancel profile editing before saving changes. The system discards unsaved modifications and restores previously saved profile information.		
Post conditions	The system updates and displays the Customer's latest profile information successfully. If no changes are saved, the original profile information remains unchanged.		
Business Rule	BR-01, BR-03, BR-04, BR-05		

- **3.1.6 UC 06: Manage Customer Accounts**

ID and Name	UC06 - Manage Customer Accounts		
Created by	Do Duc Anh		
Primary Actor	Admin	Secondary Actor	None
Description	This function allows administrators to view, search, and manage customer accounts. The administrator can monitor account statuses, inspect profile details, and enforce security policies by banning accounts that violate terms of service (e.g., fraudulent refund claims, malicious activities) or unbanning accounts after verification.		
Preconditions	1. The administrator has successfully logged into the system. 2. The administrator has access to the Account Management module within the Admin Panel.		
Main Sequence	1. Admin navigates to the "Customer Accounts" section from the admin sidebar menu. 2. System retrieves and displays a paginated list of all registered customer accounts, showing User ID, Name, Email, Registration Date, and Current Status (Active/Banned). 3. Admin inputs specific criteria (e.g., Email, Phone Number, or Account Status) in the search bar to filter the customer list. 4. Admin selects a specific customer record and clicks the "Ban Account" button due to verified policy violations. 5. System prompts a modal window requiring a reason for the ban, updates the account status to "Banned" in the database, terminates any active sessions for that user, and logs the administrative action.		
Alternative Sequences	Step 4: If the selected account is already suspended and the admin wishes to restore access, the admin clicks the "Unban Account" button. System prompts a confirmation modal, reverts the account status to "Active", and dispatches an automated notification email to the customer. Step 5: If the admin attempts to ban a system account or another administrator account, the system blocks the action, displays an error message: "Access Denied. Administrative accounts cannot be banned via this interface," and flags the event in the security logs.		
Post conditions	System successfully modifies the target account's access privileges in the database and updates the account lifecycle audit trail.		
Business Rule	BR-03, BR-40		

• **3.1.7 UC 07: Manage Manager Accounts**

ID and Name	UC07 - Manage Manager Accounts		
Created by	Do Duc Anh		
Primary Actor	Admin	Secondary Actor	None
Description	This function allows administrators to view, update, deactivate, or delete corporate Manager accounts. The administrator utilizes this feature to manage internal staff privileges, update manager profiles, and revoke system access for offboarded personnel to safeguard store operations.		
Preconditions	1. The administrator has successfully logged into the system. 2. The administrator has root/full access privileges to the Staff & Role Management module.		
Main Sequence	1. Admin navigates to the "Manager Identity & Access Control" panel within the admin dashboard. 2. System retrieves and displays a comprehensive list of all registered Manager accounts, including Staff ID, Full Name, Assigned Warehouse/Branch, Account Status (Active/Suspended), and Last Login Timestamp. 3. Admin clicks on a specific Manager account entry to view its detailed administrative profile. 4. Admin modifies the manager's access parameters (e.g., changing branch assignment, updating contact email, or toggling the account status to "Suspended"). 5. Admin clicks the "Save Changes" button to commit the modifications. 6. System updates the manager's profile record in the database, invalidates the manager's active browser tokens if suspended, and logs the administrative modification event.		
Alternative Sequences	Step 4: If a manager leaves the company permanently, the admin can click the "Delete Account" button. The system checks if the manager is currently bound to any active daily waybills (UC48) or pending orders (UC47). If no active bindings exist, the system prompts a double-confirmation modal and executes a soft-delete (marking the record as archived) to preserve historical data integrity. Step 4 (Constraint): If the manager is currently processing active shipments or pending orders, the system blocks the deletion/deactivation request and displays an error message: "Cannot modify account. This Manager is currently assigned to active fulfillment tasks. Please reassign the tasks first."		
Post conditions	System successfully alters or archives the target Manager account's credentials and authorization state in the database without disrupting historical transactional logs.		
Business Rule	BR-40		

- **3.1.8 UC 08: Manage Shipper Accounts**

ID and Name	==UC08 - Manage Shipper Accounts		
Created by	Do Duc Anh		
Primary Actor	Admin	Secondary Actor	None
Description	This function allows administrators to review registrations, view profile details, and deactivate or soft-delete accounts of delivery personnel (Shippers). This ensures only verified couriers operate within the platform and allows immediate access revocation for offboarded or non-compliant drivers.		
Preconditions	1. The administrator has successfully logged into the system. 2. The administrator has full access privileges to the Fleet Logistics & Role Management module.		
Main Sequence	1. Admin navigates to the "Shipper Account & Fleet Management" panel in the admin console. 2. System retrieves and displays a comprehensive list of all registered Shippers, including Shipper ID, Full Name, Vehicle Plate Number, Account Status (Active/Suspended/Pending Approval), and Total Completed Deliveries. 3. Admin clicks on a specific Shipper entry to view their complete background profile, including scanned driver's licenses, identity cards, and current wallet balance. 4. Admin modifies account configurations or toggles the shipper's operational status to "Suspended" due to policy infractions. 5. Admin clicks the "Save Changes" button to commit the modifications. 6. System updates the shipper's record in the centralized database, invalidates the shipper's active mobile application tokens if suspended, and logs the administrative intervention.		
Alternative Sequences	Step 3: If the shipper's profile status is "Pending Approval" (New Registration), the admin reviews the uploaded onboarding documentation and clicks either "Approve Profile" (activating the account and sending temporary credentials via SMS) or "Reject Profile" (deleting the temporary application and notifying the applicant of the rejection reason). Step 4: If the admin clicks the "Delete Account" button for an offboarded shipper, the system checks if the shipper is currently assigned to any active delivery batches (UC48) or holds an unliquidated cash-on-delivery (COD) balance. If active tasks or debts exist, the system blocks the deletion and alerts the admin: "Action Denied. Shipper has active deliveries or outstanding COD balances."		
Post conditions	System successfully modifies or archives the target Shipper's authorization state in the database, restricting or granting access to the mobile dispatch ecosystem immediately.		
Business Rule	BR-40		

3.2 Customer Order & Shopping

- **3.2.1 UC 09: Browse and Search Products**

ID and Name	UC09 - Browse & Search Product		
Created by	Trịnh Minh Quang		
Primary Actor	Guest	Secondary Actor	None
Description	Allows visitors to view the product list, search for specific fruit products using keywords, or filter products by global categories without authenticating.		
Preconditions	1 The Guest has accessed the FreshMart website homepage.		
Main Sequence	1. Guest views the default product list displayed on the homepage. 2. The guest enters a keyword (e.g., "Apple", "Mango") into the search bar. 3. The guest selects a specific product category from the filter menu. 4. Guest triggers the search/filter action. 5. The system processes the query and filters the items from the products table based on the criteria. 6. The system displays the relevant product cards matching the search/filter parameters (including name, thumbnail image, price).		
Alternative Sequences	Step 5: If no products match the keyword or selected category, the system displays a message: "No products found matching your criteria" along with suggestions to try other keywords.		
Post conditions	The filtered product catalog is rendered correctly on the user interface.		
Business Rule	BR-01, BR-20		

- **3.2.2 UC 10: View Product Details**

ID and Name	UC10 - View Product Details		
Created by	Trịnh Minh Quang		
Primary Actor	Guest	Secondary Actor	Customer
Description	Displays comprehensive product information including images, descriptions, prices, real-time stock availability, and historical user reviews.		
Preconditions	1. The user is browsing the product catalog or has executed a search query.		
Main Sequence	1. The user clicks on a specific product card from the marketplace list. 2. System queries the database tables(products, inventory, review). 3. The system calculates the aggregate user ratings and checks current stock levels. . 4. The system displays the dedicated product detail page with all retrieved data attributes. .		

Alternative Sequences	Step 2: If the chosen product has just been disabled or deleted by a Manager concurrently, the system redirects to the homepage and displays: "Product is no longer available."
Post conditions	Detailed product data specifications and user ratings are fully rendered on the screen.
Business Rule	BR-20

• **3.2.3 UC 11: Add Product to Cart**

ID and Name	UC11 - Add Product to Cart		
Created by	Trần Bảo Lâm		
Primary Actor	Customer	Secondary Actor	None
Description	Allows a logged-in Customer to add a selected fruit product with a specified quantity into the shopping cart.		
Preconditions	1. The Customer is logged into the system. 2. The Customer is viewing the Product Details page of an active product.		
Main Sequence	1. Customer adjusts the desired quantity of the selected product by clicking the “+” or “-” buttons or entering a quantity value. 2. Customer selects the “Add to Cart” option. 3. System validates the requested quantity. 4. System checks whether the requested quantity is less than or equal to the available stock quantity. 5. System adds the selected product and quantity into the Customer’s shopping cart. 6. System recalculates the shopping cart total and item quantity. 7. System confirms successful addition and updates the shopping cart information.		
Alternative Sequences	Step 4: If the requested quantity exceeds available warehouse stock, the system rejects the request and displays the message: “Cannot add item. Only [X] items left in stock.” Step 5: If the product already exists in the shopping cart and the combined quantity exceeds available stock, the system rejects the request and displays the message: “You already have [Y] items in your cart. You cannot add more than the available stock.” Step 2: Customer may cancel the add-to-cart action, and the system terminates the process without changing the shopping cart.		
Post conditions	The selected product is successfully added to the Customer’s shopping cart with the correct quantity and updated cart information.		
Business Rule	BR-01, BR-07, BR-13, BR-20		

• 3.2.4 UC 12: View Shopping Cart

ID and Name	UC12 - View Shopping Cart		
Created by	Trần Bảo Lâm		
Primary Actor	Customer	Secondary Actor	None
Description	Allows a logged-in Customer to review fruit products added to the shopping cart, view product information and prices, and check calculated subtotals and grand total.		
Preconditions	1. The Customer is logged into the system. 2. The Customer is currently on a page containing the Cart icon in the navigation header.		
Main Sequence	1. Customer selects the “Cart” icon from the navigation header. 2. System retrieves all products currently stored in the Customer’s shopping cart. 3. System retrieves product information, including product names, images, and latest prices. 4. System checks the current stock availability for each product in the shopping cart. 5. System calculates the subtotal for each product item and the grand total of the shopping cart. 6. System displays the Shopping Cart page with detailed product information and calculated totals.		
Alternative Sequences	Step 2: If the shopping cart is empty, the system displays an “Empty Cart” message and provides an option to “Return to Shop.” Step 3: If a product price has changed since being added to the shopping cart, the system updates the displayed price and recalculates shopping cart totals. Step 4: If a product is no longer active or unavailable for sale, the system marks the product as “Unavailable” and excludes it from the total price calculation.		
Post conditions	The system displays the Shopping Cart page with updated product information, latest prices, and accurate calculated totals.		
Business Rule	BR-01, BR-20, BR-21		

• 3.2.5 UC 13: Update Cart Item Quantity

ID and Name	UC13 - Update Cart Item Quantity		
Created by	Trần Bảo Lâm		
Primary Actor	Customer	Secondary Actor	None
Description	Allows a logged-in Customer to update the quantity of a selected product in the shopping cart and automatically recalculate the shopping cart totals.		
Preconditions	1. The Customer is logged into the system. 2. The Customer is actively viewing the Shopping Cart page (UC15).		
Main Sequence	1. Customer adjusts the quantity of a selected product by clicking the “+” or “-” buttons or entering a quantity value. 2. System validates the requested quantity against the available stock quantity. 3. System updates the quantity of the selected product in the Customer’s shopping cart. 4. System recalculates the product subtotal and the grand total of the shopping cart. 5. System updates the shopping cart information and displays the latest calculated values.		
Alternative Sequences	Step 1: If the Customer decreases the quantity to zero, the system forwards the process to UC17 - Remove Item from Cart and removes the selected product from the shopping cart. Step 2: If the requested quantity exceeds available stock, the system rejects the update, resets the quantity to the maximum available stock level, and displays the message: “Maximum available stock reached.”		
Post conditions	The system updates the shopping cart with the latest valid product quantity and recalculated totals.		
Business Rule	BR-01, BR-07, BR-13, BR-21		

● 3.2.6 UC 14: Remove Item from Cart

ID and Name	UC14 - Remove Item from Cart		
Created by	Trần Bảo Lâm		
Primary Actor	Customer	Secondary Actor	None
Description	Allows a logged-in Customer to remove a selected product from the shopping cart.		
Preconditions	1. The Customer is logged into the system. 2. The Customer is actively viewing the Shopping Cart page (UC15).		
Main Sequence	1. Customer selects the “Remove” option (or trash icon) for a specific product in the shopping cart. 2. System displays a confirmation dialog requesting deletion confirmation. 3. Customer confirms the deletion request. 4. System removes the selected product from the Customer’s shopping cart. 5. System recalculates the shopping cart totals based on remaining products. 6. System updates the shopping cart information and displays the latest calculated totals.		
Alternative Sequences	Step 3: Customer may cancel the deletion request. The system closes the confirmation dialog and keeps the shopping cart unchanged. Step 5: If the shopping cart becomes empty after item removal, the system displays an “Empty Cart” message and provides an option to “Return to Shop.”		
Post conditions	The selected product is removed from the shopping cart and the updated shopping cart information is displayed.		
Business Rule	BR-01, BR-21		

- **3.2.7 UC 15: View Order History**

ID and Name	UC15 - View Order History		
Created by	Đỗ Nguyễn Hùng		
Primary Actor	Customer	Secondary Actor	None
Description	Allows a logged-in Customer to review the complete history of previously placed orders, including order summary information and current order status.		
Preconditions	1. The Customer is logged into the system. 2. The Customer account is active and accessible. 3. The Customer has previously placed at least one order.		
Main Sequence	1. Customer selects the “My Orders” option from the account menu. 2. System retrieves all orders associated with the logged-in Customer. 3. System sorts the retrieved orders by creation time (newest to oldest). 4. System displays the Order History page, including order summaries such as order ID, order date, total amount, payment status, and shipping status. 5. Customer reviews the displayed order history.		
Alternative Sequences	Step 2: If no order history exists for the Customer account, the system displays the message: “No order history found.” and provides an option to “Continue Shopping.”		
Post conditions	The Customer successfully views the latest order history information.		
Business Rule	BR-01, BR-25, BR-26		

- **3.2.8 UC 16: View Order Details**

ID and Name	UC16 - View Order Details		
Created by	Đỗ Nguyễn Hùng		
Primary Actor	Customer	Secondary Actor	None
Description	Allows a logged-in Customer to view detailed information of a selected order, including purchased products, prices, quantities, payment information, shipping status, and order timeline.		
Preconditions	1. The Customer is logged into the system. 2. The Customer is currently on the Order History page (UC19). 3. At least one order exists.		
Main Sequence	1. Customer selects an order and clicks the “View Details” button. 2. System retrieves detailed information of the selected order. 3. System retrieves product information including product name, quantity, purchase price, and subtotal. 4. System retrieves payment information, shipping address, shipping method, and current order status. 5. System displays the Order Details page with full order information. 6. Customer reviews the order details.		

Alternative Sequences	Step 2: If the order cannot be found, the system displays the message: “Order not found.” Step 2: If the system detects that the order does not belong to the logged-in Customer, the system blocks access and displays the message: “Access denied.”
Post conditions	The Customer successfully views detailed information of the selected order.
Business Rule	BR-01, BR-19, BR-25

• **3.2.9 UC 17: Cancel Order**

ID and Name	UC17 - Cancel Order		
Created by	Đỗ Nguyễn Hùng		
Primary Actor	Customer	Secondary Actor	None
Description	Allows a logged-in Customer to cancel an order when the order status is still eligible for cancellation (e.g., pending or preparing).		
Preconditions	1. The Customer is logged into the system. 2. The Customer is currently on the Order Details page (UC20). 3. The selected order status allows cancellation.		
Main Sequence	1. Customer clicks the “Cancel Order” button. 2. Customer choose the reason for cancelled 3. System displays a cancellation confirmation dialog. 4. Customer confirms the cancellation request. 5. System validates whether the order is eligible for cancellation. 6. System updates the order status to “Cancelled”. 7. System updates inventory quantities for cancelled products. 8. System displays a success notification and refreshes order information.		
Alternative Sequences	Step 3: Customer may cancel the action by closing the confirmation dialog or clicking “Cancel”. The system terminates the process and keeps the order unchanged. Step 4: If the order cannot be cancelled (status is shipping, completed, or already cancelled), the system rejects the request and displays the message: “This order can no longer be cancelled.” Step 3: If the Customer does not select a cancellation reason, the system highlights the required field and displays a validation message requesting correction.		
Post conditions	The selected order is cancelled successfully (if eligible), and inventory quantities are updated accordingly.		
Business Rule	BR-01, BR-25, BR-27, BR-28, BR-29		

• 3.2.10 UC 18: Rate and Review Products

ID and Name	UC18 - Rate and Review Products		
Created by	Đỗ Nguyễn Hùng		
Primary Actor	Customer	Secondary Actor	None
Description	Allows a logged-in Customer to submit ratings and reviews for products purchased in completed orders.		
Preconditions	1. The Customer is logged into the system. 2. The Customer has successfully received the purchased order. 3. The selected product belongs to a completed order.		
Main Sequence	1. Customer selects a purchased product and clicks the “Rate & Review” option. 2. System displays the rating and review form. 3. Customer selects a rating score and enters a review comment. 4. Customer submits the review by clicking the “Submit Review” button. 5. System validates the review content and eligibility conditions. 6. System stores the product rating and review. 7. System displays a success notification and updates product review information.		
Alternative Sequences	Step 5: If the selected product does not belong to a completed order, the system rejects the request and displays the message: “Only completed purchases may be reviewed.” Step 5: If the review content is empty or the rating value is invalid, the system displays a validation error and requests correction.		
Post conditions	The system successfully records the Customer’s product rating and review.		
Business Rule	BR-01, BR-25, BR-30, BR-31, BR-32		

• 3.2.11 UC 19: View Managed Shops

ID and Name	UC19 – View Managed Shops		
Created by	Đỗ Nguyễn Hùng		
Primary Actor	Manager	Secondary Actor	None
Description	Allows a Manager to view all shops assigned to them and select a shop to manage.		
Preconditions	1. Manager is logged in. 2. Manager has at least one assigned shop.		
Main Sequence	1. Manager opens “My Shops” section. 2. System retrieves assigned shops. 3. System displays shop list. 4. Manager selects a shop. 5. System loads shop context for operations.		

Alternative Sequences	Step 2: If no shop assigned, system displays message: “No shops assigned”.
Post conditions	Manager can access selected shop context.
Business Rule	BR-07

- **3.2.12 UC 20: View Shop Details**

ID and Name	UC20 – View Shop Details		
Created by	Đỗ Nguyễn Hùng		
Primary Actor	Administrator / Manager	Secondary Actor	None
Description	Allows users to view detailed information about a shop including profile, status, and business information.		
Preconditions	1. User is logged in or has access permission. 2. Shop exists in system.		
Main Sequence	1. User selects a shop. 2. System retrieves shop details. 3. System displays shop profile information. 4. User views detailed information.		
Alternative Sequences	Step 2: If shop not found, system displays error message.		
Post conditions	Shop details are displayed successfully.		
Business Rule	BR-07		

3.3 Product Catalog Management

- **3.3.1 UC 21: View Product Catalog**

ID and Name	UC21 — View Product Catalog		
Created by	Nguyen Quang Huy		
Primary Actor	Manager	Secondary Actor	None
Description	Manager views the full list of products belonging to their shop, including pricing, category, and active status.		
Preconditions	• Manager is logged in. • Manager has at least one product registered under their shop.		
Main Sequence	1. Manager navigates to the Product Catalog section. 2. System retrieves all products associated with the manager's shop. 3. System displays product list with: product name, category, price, and is_active status.		

	4. Manager reviews the catalog.
Alternative Sequences	2a: No products exist: 1. System displays: 'No products found. Add your first product to get started.' 3a: Manager applies search or filter: 1. System filters the list by the provided criteria (name, category, or status). 2. Matching results are displayed.
Post conditions	Manager has a complete and current view of their product catalog. No changes are made.
Business Rule	BR-20

• **3.3.2 UC 22: Create New Product**

ID and Name	UC22 — Create New Product		
Created by	Nguyen Quang Huy		
Primary Actor	Manager	Secondary Actor	None
Description	Manager adds a new product to their shop by providing all required product details.		
Preconditions	• Manager is logged in. • Manager's shop is active. • At least one product category exists in the system.		
Main Sequence	1. Manager navigates to the Product Catalog and taps 'Add New Product'. 2. System displays a product creation form. 3. Manager enters: product name, description, category, unit, and initial is_active status. 4. Manager optionally uploads a product image. 5. Manager submits the form. 6. System validates all required fields. 7. System saves the new product and assigns a unique product ID. 8. System displays a success message and the new product in the catalog.		
Alternative Sequences	6a: Required field is missing: 1. System highlights the missing fields. 2. Manager must complete them before submission. 6b: Duplicate product name detected: 1. System warns: 'A product with this name already exists in your shop.' 2. Manager may choose to proceed or rename.		

Post conditions	New product is created and saved to the shop's product catalog with is_active as set by manager.
Business Rule	BR-04, BR-07, BR-20

• **3.3.3 UC 23: Update Product Catalog**

ID and Name	UC23 — Update Product Catalog		
Created by	Nguyen Quang Huy		
Primary Actor	Manager	Secondary Actor	None
Description	Manager edits an existing product's information such as name, description, price, or category.		
Preconditions	• Manager is logged in. • The product exists in the manager's shop catalog.		
Main Sequence	1. Manager selects a product from the catalog. 2. Manager taps 'Edit Product'. 3. System displays the product edit form pre-filled with current values. 4. Manager modifies desired fields (name, description, price, category, image). 5. Manager submits the changes. 6. System validates the updated fields. 7. System saves the changes and displays a success confirmation.		
Alternative Sequences	6a: Validation fails (e.g., negative price): 1. System highlights invalid fields and displays error messages. 2. Changes are not saved until corrected. 5a: Manager cancels editing: 1. System discards all unsaved changes and returns to catalog view.		
Post conditions	Product record is updated with new values. Change timestamp is recorded.		
Business Rule	BR-04, BR-07		

• **3.3.4 UC 24: Delete Product**

ID and Name	UC24 — Delete Product		
Created by	Nguyen Quang Huy		
Primary Actor	Manager	Secondary Actor	None
Description	Manager permanently removes a product from the shop catalog, provided it has no order history (including cancelled orders).		
Preconditions	• Manager is logged in. • The product exists in the catalog. • The product has no associated order history (including cancelled orders).		
Main Sequence	1. Manager selects a product from the catalog.		

	2. Manager taps 'Delete Product'. 3. System checks whether the product has any associated order history. 4. System prompts manager to confirm deletion: 'Are you sure? This action cannot be undone.' 5. Manager confirms deletion. 6. System permanently removes the product from the database. 7. System displays a success message and updates the catalog.
Alternative Sequences	3a: Product has existing order history (including cancelled): 1. System blocks deletion and displays: 'This product cannot be deleted because it is associated with existing orders.' 2. System suggests using Toggle Availability (UC40) to hide the product instead. 5a: Manager cancels deletion prompt: 1. System discards the action and returns to catalog view.
Post conditions	Product is permanently removed from the catalog and is no longer visible to customers.
Business Rule	BR-20

• **3.3.5 UC 25: Toggle Product Availability Status**

ID and Name	UC25 — Toggle Product Availability Status		
Created by	Nguyen Quang Huy		
Primary Actor	Manager	Secondary Actor	None
Description	Manager quickly activates or deactivates a product's visibility on the storefront by toggling its <code>is_active</code> flag.		
Preconditions	• Manager is logged in. • The product exists in the catalog.		
Main Sequence	1. Manager navigates to the Product Catalog. 2. Manager locates the target product. 3. Manager taps the availability toggle switch next to the product. 4. System flips the <code>is_active</code> value (<code>true</code> → <code>false</code> or <code>false</code> → <code>true</code>). 5. System immediately reflects the new status in the catalog view. 6. System displays a brief confirmation toast: 'Product visibility updated.'		
Alternative Sequences	4a: System fails to update due to a server error: 1. System displays an error message. 2. Toggle reverts to previous state. 3. Manager is advised to try again.		
Post conditions	Product <code>is_active</code> flag is updated. If deactivated, product is hidden from all customer-facing views immediately.		

Business Rule	BR-20
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3.4 Batch Management

- **3.4.1 UC 26: View Batch List**

ID and Name	UC26 — View Batch List		
Created by	Nguyen Quang Huy		
Primary Actor	Manager	Secondary Actor	None
Description	Manager view the list of imported product batches in the system. The batch list helps the Manager monitor stock, price, expiration date, supplier, and batch status.		
Preconditions	Manager is logged in.		
Main Sequence	1. The Manager opens the Batch Management page. 2. The system retrieves all product batches. 3. The system displays the batch list with key information: Fruit name, Current stock, Selling price, Expiration date, Supplier, Batch status		
Alternative Sequences	AS1: No batch exists. The system displays an empty state message: “No batch has been created.”		
Post conditions	The Manager can view existing batch information		
Business Rule	BR-07, BR-15		

• 3.4.2. UC 27: Create Product Batch

ID and Name	UC27 — Create Product Batch		
Created by	Nguyen Quang Huy		
Primary Actor	Manager	Secondary Actor	None
Description	This use case allows the Manager to create a new imported fruit batch. After creation, the system automatically updates inventory and product price.		
Preconditions	• The Manager has logged into the system. • The product catalog contains at least one active product. • The selected product already exists in the product catalog.		
Main Sequence	1. The Manager opens the Batch Management page. 2. The Manager selects Create New Batch. 3. The system displays the batch creation form. 4. The Manager selects a fruit product from the product catalog. 5. The Manager enters batch information: Quantity, Unit, Import price, Selling price, Expiration date, Supplier name, Batch note (if any) 6. The system automatically generates the import date and time. 7. The Manager submits the form. 8. The system validates the batch information. 9. The system creates the new batch. 10. The system automatically increases the product inventory based on the batch quantity. 11. The system records the batch creation history. 12. The system displays a success message.		
Alternative Sequences	AS1: Product catalog is empty 1. The system detects that no active product exists. 2. The system displays: “No product is available. Please create a product first.” AS2: Invalid quantity 1. The Manager enters a quantity less than or equal to 0. 2. The system displays: “Quantity must be greater than 0.” 3. The Manager corrects the quantity. 4. The use case continues from Step 7 of the Main Sequence. AS3: Invalid price 1. The Manager enters an import price or selling price less than 0. 2. The system displays: “Price cannot be negative.” 3. The Manager corrects the price. 4. The use case continues from Step 7 of the Main Sequence. AS4: Invalid expiration date 1. The Manager enters an expiration date earlier than or equal to the import date. 2. The system displays: “Expiration date must be later than the import date.”		

	3. The Manager corrects the expiration date. 4. The use case continues from Step 7 of the Main Sequence. AS5: System fails to create batch 1. The system fails to save the batch due to a server or database error. 2. The system displays: “Unable to create a batch. Please try again later.”
Post conditions	• A new batch is created. • Product inventory is increased. • Product selling price is updated. • Batch creation history is saved.
Business Rule	BR-15

• **3.4.3 UC 28: View Batch Detail**

ID and Name	UC28 — View Batch Detail		
Created by	Nguyen Quang Huy		
Primary Actor	Manager	Secondary Actor	None
Description	Manager to view detailed information of a specific product batch, including batch data, inventory information, price information, and related history.		
Preconditions	• The selected batch exists in the system.		
Main Sequence	1. The Manager opens the Batch Management page. 2. The Manager selects a batch from the list. 3. The system retrieves the batch details. 4. The system displays detailed batch information: Batch ID, Fruit name, Quantity, Current stock, Import price, Selling price, Import date, Expiration date, Supplier, Batch status, Note		
Alternative Sequences	AS1: Batch does not exist 1. The system displays: “Batch not found.”		
Post conditions	• The Manager can view detailed batch information. • No batch data is changed.		
Business Rule	BR-15		

• **3.4.4 UC 29: Update Stock Quantity**

ID and Name	UC29 — View Batch Detail		
Created by	Nguyen Quang Huy		
Primary Actor	Manager	Secondary Actor	None
Description	Manager manually update the stock quantity of a product batch. Every stock change must be saved in stock history for tracking purposes.		
Preconditions	• The selected batch exists in the system. • The batch is not cancelled.		

Main Sequence	1. The Manager opens the batch detail page. 2. The Manager selects Update Stock. 3. The system displays the current stock quantity. 4. The Manager enters the new stock quantity or adjustment quantity. 5. The Manager enters the reason for the stock change. 6. The Manager submits the update. 7. The system validates the new stock value. 8. The system updates the batch stock quantity. 9. The system updates the related product inventory. 10. The system records stock change history, including: Previous stock, New stock, Changed quantity, Reason, Updated by, Updated time 11. The system displays a success message.
Alternative Sequences	AS1: Invalid stock value 1. The Manager enters a stock value less than 0. 2. The system displays: "Stock quantity cannot be negative." 3. The Manager corrects the stock value. 4. The use case continues from Step 6 of the Main Sequence. AS2: Missing update reason 1. The Manager does not enter the reason for stock change. 2. The system displays: "Reason is required." 3. The Manager enters the reason. 4. The use case continues from Step 6 of the Main Sequence. AS3: Batch is cancelled 1. The Manager selects Update Stock for a cancelled batch. 2. The system displays: "Cancelled batch cannot be updated."
Post conditions	● Batch stock quantity is updated. ● Related product inventory is updated. ● Stock change history is recorded.
Business Rule	BR-15

● **3.4.5 UC 30: Update Selling Price**

ID and Name	UC30 — Update Selling Price		
Created by	Nguyen Quang Huy		
Primary Actor	Manager	Secondary Actor	None
Description	This use case allows the Manager to update the selling price of a product batch or product. Every price change must be saved in price history		
Preconditions	● The selected batch or product exists in the system.		
Main Sequence	1. The Manager opens the batch detail page. 2. The Manager selects Update Price. 3. The system displays the current selling price. 4. The Manager enters the new selling price.		

	- The Manager enters the reason for the price change. - The Manager submits the update. - The system validates the new selling price. - The system updates the selling price. - The system records price change history, including: Previous price, New price, Reason, Updated by, Updated time - The system displays a success message.
Alternative Sequences	AS1: Invalid selling price - The Manager enters a selling price less than 0. - The system displays: "Selling price cannot be negative." - The Manager corrects the selling price. - The use case continues from Step 6 of the Main Sequence. AS2: Missing update reason - The Manager does not enter the reason for price change. - The system displays: "Reason is required." - The Manager enters the reason. - The use case continues from Step 6 of the Main Sequence.
Post conditions	- Selling price is updated. - Price change history is recorded. - The updated price is used for future sales.
Business Rule	BR-15

• **3.4.6 UC 31: View Stock Change History**

ID and Name	UC31 — View Stock Change History		
Created by	Nguyen Quang Huy		
Primary Actor	Manager	Secondary Actor	None
Description	This use case allows the Manager to view the stock change history of a batch or product.		
Preconditions	The selected batch or product exists in the system.		
Main Sequence	- The Manager opens the batch detail page. - The Manager selects View Stock History. - The system retrieves stock change history. - The system displays stock history information: Previous stock, New stock, Changed quantity, Reason, Updated by, Updated time		
Alternative Sequences	AS1: No stock history exists The system displays: "No stock history available."		
Post conditions	- The Manager can view stock change history. - No data is changed.		

Business Rule	BR-15

• **3.4.7 UC 32: View Price Change History**

ID and Name	UC32 — View Price Change History		
Created by	Nguyen Quang Huy		
Primary Actor	Manager	Secondary Actor	None
Description	This use case allows the Manager to view the price change history of a batch or product.		
Preconditions	The selected batch or product exists in the system.		
Main Sequence	5. The Manager opens the batch detail page. 6. The Manager selects View Price History. 7. The system retrieves price change history. 8. The system displays price history information: Previous price, New price, Reason, Updated by, Updated time		
Alternative Sequences	AS1: No price history exists The system displays: “No price history available.”		
Post conditions	- The Manager can view price change history. - No data is changed.		
Business Rule	BR-15		

3.5 Customer Account & Transaction Services

- 3.5.1 UC 33: *Checkout and Place Order*

ID and Name	UC33 - Checkout and Place Order		
Created by	Trần Bảo Lâm		
Primary Actor	Customer	Secondary Actor	Online Payment Gateway (e.g., VNPAY)
Description	Allows a logged-in Customer to review shopping cart items, select or add a delivery address, choose a shipping method, select a payment method (Cash on Delivery or Online Payment), and place an order. The system calculates final costs, updates inventory, and generates order records.		
Preconditions	- The Customer is logged into the system. - The Customer is actively viewing the Shopping Cart page (UC15). - The shopping cart contains at least one valid product.		
Main Sequence	- Customer clicks the “Proceed to Checkout” button from the Shopping Cart page. - System validates that all cart items are active and available in sufficient stock. - System displays the checkout page with default delivery address, available shipping methods, and payment options. - Customer selects or updates delivery address, chooses shipping method, and selects a payment method. - System calculates shipping fee and updates the final total (product total + shipping fee - discounts if any). - Customer confirms the order by clicking “Place Order”. - System performs final stock validation and temporarily reserves inventory. - System creates a new order with status “Pending” (or “Paid” if online payment is successful). - System updates inventory by deducting purchased quantities. - System clears all items from the Customer’s shopping cart. - System displays order confirmation.		
Alternative Sequences	Step 2: If one or more cart items are no longer available or out of stock, the system stops the checkout process and redirects the Customer back to the cart page with the message: “Some items in your cart are no longer available.” Step 4: Customer may add a new delivery address during checkout. The system validates and saves the new address to the Customer profile and recalculates shipping fee. Step 6: If Customer selects Online Payment, the system redirects to the Payment Gateway for transaction processing. After successful payment, the system updates the order status to “Paid”.		

	Step 6-E1: If payment fails or is cancelled, the system marks the order as “Payment Failed”, releases reserved stock, and notifies the Customer. Step 7: If stock becomes unavailable during checkout (race condition), the system cancels order creation and notifies the Customer of the unavailable product.
Post conditions	1. A new order is created successfully with status Pending or Paid. 2. Inventory quantities are updated accordingly. 3. The Customer’s shopping cart is cleared.
Business Rule	BR-01, BR-07, BR-12, BR-14, BR-15, BR-16, BR-17, BR-18, BR-19, BR-22, BR-23

• **3.5.2 UC 34: Request Refund / Return**

ID and Name	UC34 - Request Refund / Return		
Created by	Đỗ Nguyễn Hùng		
Primary Actor	Customer	Secondary Actor	None
Description	Allows a logged-in Customer to submit a refund or return request for purchased products by selecting affected items, providing reasons, uploading supporting evidence, and choosing refund and return methods.		
Preconditions	1. The Customer is logged into the system. 2. The Customer has a completed order eligible for refund or return. 3. The refund request period has not expired.		
Main Sequence	1. Customer selects an eligible order and clicks the “Request Refund / Return” option. 2. System retrieves purchased products eligible for refund or return. 3. Customer selects affected product(s). 4. Customer enters the refund reason and uploads supporting evidence (e.g., damaged product image). 5. Customer selects a return method and refund method. 6. Customer submits the request by clicking the “Submit Request” button. 7. System validates the request information and eligibility conditions. 8. System creates a refund request record with status “Pending”. 9. System displays a success notification and updates refund request information.		
Alternative Sequences	Step 2: If the refund period has expired, the system rejects the request and displays the message: “Refund period expired.” Step 4: If required information is missing (reason, method, or evidence when required), the system displays a validation message requesting completion. Step 7: If a duplicate refund request exists for the same order/product, the system rejects the request and displays a warning message.		

Post conditions	The system records a refund or return request successfully with Pending status for manager review.
Business Rule	BR-01, BR-25, BR-33, BR-34, BR-35, BR-36

• **3.5.3 UC 35: View Notifications**

ID and Name	UC35 - View Notifications		
Created by	Trần Bảo Lâm		
Primary Actor	Customer	Secondary Actor	None
Description	Allows a logged-in Customer to view real-time and historical notifications regarding order updates, security alerts, and marketing promotions.		
Preconditions	1. The Customer is successfully logged into the system.		
Main Sequence	1. Customer clicks the “Notification” (bell icon) in the navigation header. 2. System retrieves the list of notifications for the Customer, sorted from newest to oldest. 3. System displays a dropdown list of recent notifications and highlights unread items. 4. Customer selects “View All Notifications”. 5. System redirects the Customer to the Notification Center page displaying all notifications categorized by type. 6. Customer selects a specific notification to view details. 7. System displays the full content of the selected notification and marks it as “Read”. 8. System updates the unread notification counter dynamically.		
Alternative Sequences	Step 2: If no notifications exist, the system displays the message: “You have no notifications at this time.” Step 6: If the notification is actionable (e.g., order update), the system marks it as read and redirects the Customer to the relevant page (e.g., Order Details page).		
Post conditions	The system displays updated notifications, updates unread counters, and marks viewed notifications as read.		
Business Rule	BR-37, BR-38		

• **3.5.4 UC 36: View Shop Information**

ID and Name	UC36 - View Shop Information		
Created by	Trịnh Minh Quang		
Primary Actor	Guest	Secondary Actor	Customer
Description	Allows users to view the comprehensive profile, business background, and designated storefront details of a specific vendor/store on the platform.		

Preconditions	1. The user is viewing a product page or store directory listing.
Main Sequence	1. Guest clicks on the shop's name or vendor profile link. 2. The system retrieves vendor details from the shop database table. 3. The system displays the shop's profile page containing its name, introduction, contact address, policies, and its specific active product list
Alternative Sequences	Step 2: If the target shop account is currently banned or inactive, the system displays: "This shop is temporarily unavailable" and terminates the process.
Post conditions	The designated shop information screen is successfully displayed to the user.
Business Rule	BR-01

- **3.5.5 UC 37: View Website Information**

ID and Name	UC37 - View Website Information		
Created by	Trịnh Minh Quang		
Primary Actor	Guest	Secondary Actor	Customer

Description	Allows any public visitor to view static system content including the platform's introduction, operational policies, and terms of service.		
Preconditions	1. The user has accessed the platform website.		
Main Sequence	1. Guest navigates to the footer or information menu and selects a topic (e.g., "Terms of Service", "Privacy Policy"). 2. System fetches the requested static content details. 3. System renders the corresponding information document view on the interface.		
Alternative Sequences	None.		
Post conditions	The requested corporate info or marketplace policy text is successfully presented on the screen.		
Business Rule	None		

- **3.5.6 UC 38: View Address Book**

ID and Name	UC38 - View Address Book		
Created by	Đỗ Nguyễn Hùng		
Primary Actor	Customer	Secondary Actor	None

Description	Allows a logged-in Customer to view the list of saved delivery addresses, identify the default address, and review address details before selecting, updating, or deleting an address.
Preconditions	1. The Customer is logged into the system. 2. The Customer account is active. 3. The Customer has accessed the Address Book section from the account page.
Main Sequence	1. Customer selects the “Address Book” option from the account menu. 2. System retrieves all saved addresses belonging to the Customer. 3. System identifies the default delivery address. 4. System displays the Address Book page with all saved addresses, including receiver name, phone number, province, district, ward, and detailed address. 5. System highlights the default delivery address.
Alternative Sequences	Step 2: If no saved addresses exist, the system displays the message: “ No saved address found. ” and provides an option to “ Add New Address. ”
Post conditions	The system displays the Customer’s saved delivery addresses and clearly identifies the default address.
Business Rule	BR-01, BR-03, BR-08, BR-09

● **3.5.7 UC 39: Add New Address**

ID and Name	UC39 - Add New Address		
Created by	Đỗ Nguyễn Hùng		
Primary Actor	Customer	Secondary Actor	None
Description	Allows a logged-in Customer to add a new delivery address, including receiver information, phone number, location details, and optionally set it as the default delivery address.		
Preconditions	1. The Customer is logged into the system. 2. The Customer is currently on the Address Book page (UC09).		
Main Sequence	1. Customer selects the “Add New Address” button. 2. System displays the address input form. 3. Customer enters address information (receiver name, phone number, province, district, ward, detailed address). 4. Customer optionally selects the “Set as Default Address” option. 5. Customer submits the form by clicking the “Save Address” button. 6. System validates the entered information. 7. System saves the new delivery address. 8. If selected, system sets the newly created address as the default delivery address. 9. System displays a success notification and refreshes the Address Book page.		

Alternative Sequences	Step 6: If required fields are missing, the system highlights invalid fields and displays an error message requesting completion. Step 6: If the phone number format is invalid, the system rejects the request and displays the message: “Invalid phone number format.”
Post conditions	The system successfully saves a new delivery address and updates the default address if applicable.
Business Rule	BR-01, BR-04, BR-05, BR-08, BR-09

• **3.5.8 UC 40: Update Address Details**

ID and Name	UC40 - Update Address Details		
Created by	Đỗ Nguyễn Hùng		
Primary Actor	Customer	Secondary Actor	None
Description	Allows a logged-in Customer to modify existing delivery address information or change the default delivery address.		
Preconditions	1. The Customer is logged into the system. 2. The Customer is currently on the Address Book page (UC09). 3. At least one saved address exists.		
Main Sequence	1. Customer selects an existing address and clicks “Edit Address”. 2. System retrieves the selected address information and displays it in an editable form. 3. Customer updates address information or toggles the “Set as Default Address” option. 4. Customer submits the changes by clicking “Save Changes”. 5. System validates the updated information. 6. System updates the selected address information in the database. 7. If set as default, system updates the selected address as the default delivery address and removes the default status from the previous address. 8. System displays a success notification and refreshes the Address Book page.		
Alternative Sequences	Step 5: If the input information is invalid or incomplete, the system displays an error message and requests correction. Step 4: Customer may cancel the editing process by clicking “Cancel”. The system discards all unsaved changes and restores the original address information.		
Post conditions	The selected delivery address is updated successfully.		

Business Rule	BR-01, BR-04, BR-05, BR-09, BR-11
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- **3.5.9 UC 41: Delete Address**

ID and Name	UC41 - Delete Address		
Created by	Đỗ Nguyễn Hùng		
Primary Actor	Customer	Secondary Actor	None
Description	Allows a logged-in Customer to permanently remove a saved delivery address from their account.		
Preconditions	1. The Customer is logged into the system. 2. The Customer is currently on the Address Book page (UC09). 3. At least one saved address exists.		
Main Sequence	1. Customer selects an address and clicks the “Delete Address” button. 2. System displays a confirmation dialog requesting deletion confirmation. 3. Customer confirms the deletion request. 4. System verifies ownership of the selected address. 5. System removes the selected address from the Customer account. 6. If the deleted address is the default address, system automatically assigns another available address as the new default (if any exists). 7. System refreshes the Address Book page and displays a success notification.		
Alternative Sequences	Step 3: Customer may cancel the deletion request. The system closes the dialog and keeps the address unchanged. Step 5: If the system cannot locate the selected address, it displays the message: “Address not found.”		
Post conditions	The selected delivery address is permanently removed from the Customer account and the address list is updated accordingly.		
Business Rule	BR-01, BR-09, BR-10, BR-11		

3.6 Logistics & Order Fulfillment Management

- **3.6.1 UC 42: View Delivery History**

ID and Name	UC42 — View Delivery History
Created by	Nguyen Quang Huy

Primary Actor	Shipper	Secondary Actor	None
Description	Allows the shipper to review past delivery records, filtered by date or month, showing both successful and failed deliveries.		
Preconditions	- Shipper is logged into the system. - At least one delivery has been completed or attempted.		
Main Sequence	- Shipper navigates to the Delivery History section. - System displays a list of all past deliveries with date, status, and order ID. - Shipper selects a filter (by day or by month). - System retrieves and displays matching delivery records. - Shipper views the details of each delivery entry including success/failure status.		
Alternative Sequences	4a: No records found for selected filter: - System displays a message: 'No delivery records found for the selected period.' 3a: Shipper clears filter: - System resets and shows all available delivery history.		
Post conditions	Delivery history is displayed correctly based on applied filters. No data is modified.		
Business Rule	BR-01		

• **3.5.2 UC 43: View Pending Delivery List**

ID and Name	UC43 — View Pending Delivery List		
Created by	Nguyen Quang Huy		
Primary Actor	Shipper	Secondary Actor	None
Description	Shipper browses the list of orders awaiting pickup and delivery in their assigned area.		
Preconditions	- Shipper is logged in. - Shipper has an assigned delivery area. - There are unassigned orders in the system.		
Main Sequence	- Shipper navigates to the Pending Deliveries section. - System retrieves all unassigned orders within the shipper's delivery area. - System displays a list showing order ID, destination district, item count, and COD amount. - Shipper scrolls through the list to review available orders.		
Alternative Sequences	2a: No pending orders in the area: System displays: 'No orders available in your area at this time.'		
Post conditions	Shipper has viewed all available pending orders. No orders are assigned yet.		
Business Rule	BR-01		

• **3.6.3 UC 44: Accept Delivery Order**

ID and Name	UC44 — Accept Delivery Order		
Created by	Nguyen Quang Huy		
Primary Actor	Shipper	Secondary Actor	None
Description	Shipper selects and accepts one or more pending orders from the list to take on for delivery.		
Preconditions	- Shipper is logged in. - Shipper is viewing the pending delivery list (UC26). - Selected order is still unassigned.		

Main Sequence	1. Shipper selects an order from the pending list. 2. System displays a brief order summary (destination, COD amount, item count). 3. Shipper taps 'Accept Order'. 4. System assigns the order to the shipper and updates its status to 'Picked Up - Pending'. 5. System confirms acceptance with a notification.
Alternative Sequences	3a: Order already accepted by another shipper: 1. System displays: 'This order has just been taken. Please choose another.' 2. System refreshes the pending list.
Post conditions	Order is assigned to the shipper with status updated. Order no longer appears in other shippers' pending lists.
Business Rule	BR-01, BR-24

- **3.6.4 UC 45: View Delivery Specifications**

ID and Name	UC45 — View Delivery Specifications		
Created by	Nguyen Quang Huy		
Primary Actor	Shipper	Secondary Actor	None
Description	Shipper views the full delivery details for an accepted order, including recipient info, address, COD amount, and tracking code.		
Preconditions	• Shipper is logged in. • Shipper has accepted at least one delivery order.		
Main Sequence	1. Shipper navigates to their active orders list. 2. Shipper selects an order. 3. System displays delivery specification: recipient name, phone number, delivery address, COD amount to collect, and tracking code. 4. Shipper reviews all details before proceeding to deliver.		
Alternative Sequences	3a: Order was cancelled by customer after acceptance: 1. System flags the order as cancelled. 2. System notifies shipper that the order is no longer valid..		
Post conditions	Shipper has all required information to perform the delivery.		
Business Rule	BR-01, BR-24		

- **3.6.5 UC 46: Update Order Delivery Status**

ID and Name	UC46 — Update Order Delivery Status		
Created by	Nguyen Quang Huy		
Primary Actor	Shipper	Secondary Actor	None
Description	Shipper manually updates the delivery status of an order across any of the defined milestones.		
Preconditions	• Shipper is logged in. • Shipper has an order assigned to them.		
Main Sequence	1. Shipper opens the order detail screen. 2. Shipper taps 'Update Status'. 3. System displays available status options: Not Yet Picked Up, In Transit, Delivered, Delivery Failed. 4. Shipper selects the appropriate status. 5. System saves the new status with a timestamp. 6. System notifies the customer of the status change.		
Alternative Sequences	4a: Shipper selects 'Delivered':		

	1. System prompts shipper to upload proof of delivery (links to UC30). 2. If no proof uploaded, system still saves status but flags it for review. 4b: Shipper selects 'Delivery Failed': 1. System prompts shipper to record failure reason (links to UC31).
Post conditions	Order status is updated and timestamped. Customer is notified. Status history is logged.
Business Rule	BR-01, BR-24

- **3.6.6 UC 47: Upload Delivery Proof**

ID and Name	UC47 — Upload Delivery Proof		
Created by	Nguyen Quang Huy		
Primary Actor	Shipper	Secondary Actor	None
Description	Shipper captures and uploads a photo as evidence that a delivery was successfully completed.		
Preconditions	• Shipper is logged in. • The order status has been set to 'Delivered'. • Shipper has access to their device camera.		
Main Sequence	1. Shipper opens the order that has status 'Delivered'. 2. Shipper taps 'Upload Proof of Delivery'. 3. System opens camera or file picker. 4. Shipper captures or selects a photo. 5. System validates the image (format and size). 6. Shipper confirms the upload. 7. System stores the image linked to the order and marks proof as submitted.		
Alternative Sequences	5a: Image file exceeds size limit or invalid format: 1. System displays an error message. 2. Shipper is prompted to re-capture or choose a different file. 3a: Order status is not 'Delivered': 1. System disables the upload button and shows: 'Proof can only be uploaded for delivered orders.'		
Post conditions	Delivery proof image is stored and linked to the order. The order is marked as verified.		
Business Rule	BR-01		

- **3.6.7 UC 48: Record Delivery Failure Reason**

ID and Name	UC48 — Record Delivery Failure Reason		
Created by	Nguyen Quang Huy		
Primary Actor	Shipper	Secondary Actor	None
Description	Shipper records the reason for a failed delivery and may edit it once if needed.		
Preconditions	• Shipper is logged in. • The order status has been set to 'Delivery Failed'.		
Main Sequence	8. System prompts shipper to enter a reason for delivery failure. 9. Shipper selects a predefined reason (e.g., Customer not home, Wrong address, Refused delivery) or enters a custom note. 10. Shipper submits the reason.		

	11. System saves the failure reason linked to the order and logs the submission timestamp.
Alternative Sequences	4a: Shipper wants to update the reason (first edit): 1. Shipper navigates to the failed order. 2. Shipper taps 'Edit Reason'. 3. System allows editing if it is the first edit. 4. Shipper submits the updated reason. 5. System saves the new reason and logs the edit timestamp. 4b: Shipper attempts a second edit: 1. System displays: 'Reason can only be edited once. Please contact your supervisor if further changes are needed.' 2. Edit is blocked.
Post conditions	Failure reason is saved and linked to the order. A maximum of one edit is allowed per order.
Business Rule	BR-01, BR-04

● **3.6.8. UC 49: Process Pending Orders**

ID and Name	UC49 - Process Pending Orders		
Created by	Do Duc Anh		
Primary Actor	Manager	Secondary Actor	None
Description	This function allows managers to review the list of newly placed fruit orders that are waiting for approval. The manager can inspect order details, verify real-time warehouse stock availability, and either accept the order to initiate fulfillment or reject it with a specific reason.		
Preconditions	● The manager has successfully logged into the system. ● The manager has access to the Order Management module.		
Main Sequence	6. Manager navigates to the "Pending Orders" tab within the order management panel. 7. System retrieves and displays a list of all orders currently in the "Pending" status, sorted by placement time. 8. Manager selects a specific order to inspect its details, including customer information, items ordered, total amount, payment status, and current inventory levels for those items. 9. Manager clicks the "Accept" button to approve the order. 10. System updates the order status to "Preparing" (Sent to packaging), reserves the inventory items, and triggers a notification to the customer.		
Alternative Sequences	Step 4: If the manager discovers that a fruit item is out of stock or damaged upon physical inspection, the manager clicks the "Reject" button instead. The system prompts a pop-up requiring a rejection reason (e.g., "Out of stock," "Quality issue"), updates the order status to "Canceled," releases any held stock, and notifies the customer with the explanation. Step 5: If the customer selected an online payment method (Bank/Payment Gateway) but the payment status is still unverified or failed, the system displays a warning badge next to the order and disables the "Accept" button until the payment confirmation is received.		

Post conditions	System successfully updates the order state in the database and advances the fulfillment workflow without causing inventory discrepancies.
Business Rule	BR-18, BR-19, BR-24

● **3.5.9 UC 50: Manage Daily Waybills / Shipments**

ID and Name	RA UC48 - Manage Daily Waybills / Shipments		
Created by	Do Duc Anh		
Primary Actor	Manager	Secondary Actor	None
Description	This function allows managers to compile, organize, and manage daily waybills (shipments) generated from accepted fruit orders. The manager can aggregate multiple orders into delivery batches based on geographical routes, print physical shipping labels, and hand over the shipments to available shippers.		
Preconditions	1. The manager has successfully logged into the system. 2. The manager has access to the Order & Shipment Management module. 3. There are orders currently in the "Preparing" status (accepted by the manager but not yet picked up by shippers).		
Main Sequence	1. Manager navigates to the "Daily Waybills & Dispatch" section from the management panel. 2. System retrieves and displays all active shipments for the day, categorized by geographical zones (e.g., districts, routes) and current fulfillment status. 3. Manager selects a group of pending shipments bound for the same delivery zone and clicks the "Create Dispatch Batch" button. 4. System automatically groups the selected orders, generates a unique Master Waybill ID, and updates the order statuses to "Ready for Pickup". 5. Manager assigns the generated shipment batch to an available Shipper or opens the batch for open-pool shipper claims, then prints the corresponding delivery receipts and itemized packing slips.		
Alternative Sequences	Step 3: If a specific shipment contains highly perishable fruits (e.g., ripe strawberries, cut melons) that require immediate delivery, the manager can flag the waybill as "Express/High Priority", moving it to the top of the dispatch queue regardless of the standard batching route. Step 5: If the assigned shipper rejects the batch or fails to check-in at the warehouse within the designated pickup window, the manager clicks "Unassign Shipper", reverting the batch status back to "Ready for Pickup" so it can be reassigned to another driver.		
Post conditions	System successfully packages individual orders into operational shipping waybills, updates the dispatch logs, and alerts the designated shippers via the system.		
Business Rule	BR-24		

● **3.5.10 UC 51: Inspect Order Processing Details**

ID and Name	RA UC49 - Inspect Order Processing Details
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Created by	Do Duc Anh		
Primary Actor	Manager	Secondary Actor	None
Description	This function allows managers to view the granular fulfillment details of a specific active order within the day. The manager reviews customer data, the itemized fruit checklist, and payment verification status to ensure accurate packaging, quality control, and preparation before physical handover to the logistics team.		
Preconditions	1. The manager has successfully logged into the system. 2. The manager has access to the Order Management module. 3. The selected order is currently in an active state (e.g., "Preparing").		
Main Sequence	1. Manager accesses the active dispatch queue within the order management panel. 2. System displays a summary list of orders currently being packaged or prepared. 3. Manager clicks on a specific Order ID to inspect its comprehensive fulfillment workspace. 4. System retrieves and displays the full processing profile, including: • Customer Credentials: Full name, verified delivery address, contact phone number. • Itemized Fruit Checklist: Specific fruit names, variants (e.g., weight/size options), quantities ordered, and corresponding warehouse batch location numbers. • Financial Matrix: Total item cost, calculated shipping fees, applied coupon codes, total payment due, and live payment status (e.g., "Paid via Bank", "Pending COD"). 5. Manager verifies all parameters, completes the physical packaging of the fruit box, and marks the item checklist as verified.		
Alternative Sequences	Step 4: If the customer leaves specific custom delivery or packaging instructions (e.g., "Pack in individual gift boxes", "Deliver after 5 PM"), the system highlights these notes in a flashing orange warning box at the top of the details view to prevent fulfillment errors. Step 5: If a physical discrepancy is found during the packaging process (e.g., the last remaining batch of mangoes fails quality inspection), the manager clicks the "Flag Packing Issue" button. The system pauses the fulfillment flow, places the order on "Hold" status, and logs the internal deficit note.		
Post conditions	System records that the order details have been audited and verified for packing, preparing the transaction for waybill generation without modifying core financial records.		
Business Rule	BR-19, BR-24		

• **3.5.11 UC 52: Update Internal Order Progress**

ID and Name	RA_UC50 - Update Internal Order Progress		
Created by	Do Duc Anh		
Primary Actor	Manager	Secondary Actor	None
Description	This function allows managers to manually or systematically transition the internal fulfillment status of an order from "Preparing" (Packaging		

	completed) to "Sent" (Dispatched/Handed over to the shipper). This action officially moves the order into the logistics pipeline and triggers real-time transit notifications to the customer.
Preconditions	1. The manager has successfully logged into the system. 2. The manager has access to the Order Management module. 3. The targeted order is currently in the "Preparing" status and physical packaging has been completed (RA_UC49).
Main Sequence	1. Manager logs into the admin dashboard and navigates to the "Order Processing Pipeline" interface. 2. System displays a Kanban board or list view of all active orders segmented by their current internal sub-statuses. 3. Manager selects the specific order(s) that have been successfully boxed and hands them over to the courier. 4. Manager clicks the "Mark as Sent" button (or drags the order card from the "Preparing" column to the "Sent" column). 5. System updates the order status to "Sent" (In Transit) in the centralized database, logs the handover timestamp, and dispatches an automated push notification/email to the customer with tracking details.
Alternative Sequences	Step 4: If the system detects that the order is assigned to a specific dispatch batch (RA_UC48), the manager can click "Mark Batch as Sent". The system then automatically bulk-updates all associated orders within that batch to "Sent" status simultaneously to optimize operations. Step 5: If a network timeout or database locking error occurs during the update, the system halts the status transition, rolls back the transaction, and displays an alert: "Status update failed due to a database conflict. Please refresh and try again."
Post conditions	System successfully transitions the order state to "Sent", archiving the processing lifecycle logs and enabling live shipping tracking for the end-user.
Business Rule	BR-24

● **3.5.12 UC 53: Moderate Refund Claims**

ID and Name	RA_UC51 - Moderate Refund Claims		
Created by	Do Duc Anh		
Primary Actor	Manager	Secondary Actor	None
Description	This function allows managers to review, audit, and adjudicate refund claims submitted by customers regarding damaged, spoiled, or incorrect fruit deliveries. The manager inspects the customer's textual claim, uploaded photographic/video evidence, and order history to either approve the refund (initiating bank payback) or deny the request with a justified reason.		
Preconditions	1. The manager has successfully logged into the system. 2. The manager has access to the Order Management & Customer Support module. 3. A customer has already submitted a formal refund request within the eligible timeframe.		

Main Sequence	1. Manager navigates to the "Refund Requests & Claims" queue on the dashboard. 2. System retrieves and displays all pending refund tickets, sorted by urgency and submission timestamp. 3. Manager selects a specific ticket to open the Refund Mediation Workspace, displaying the original invoice, customer's statement, and uploaded media evidence (photos/videos of the fruit). 4. Manager clicks the "Approve Refund" button after verifying that the fruit quality or delivery mismatch was the store's fault. 5. System prompts a confirmation modal, calculates the refund amount, updates the ticket status to "Refund Approved", and communicates with the Bank/Payment Gateway system to reverse the transaction.
Alternative Sequences	Step 4: If the system detects that the order is assigned to a specific dispatch batch (RA_UC48), the manager can click "Mark Batch as Sent". The system then automatically bulk-updates all associated orders within that batch to "Sent" status simultaneously to optimize operations. Step 5: If a network timeout or database locking error occurs during the update, the system halts the status transition, rolls back the transaction, and displays an alert: "Status update failed due to a database conflict. Please refresh and try again."
Post conditions	System successfully transitions the order state to "Sent", archiving the processing lifecycle logs and enabling live shipping tracking for the end-user.
Business Rule	BR-33, BR-34, BR-35, BR-36

3.7 Analytics, Reporting & Administration

- **3.7.1 UC 54: View Revenue Overview Dashboard**

ID and Name	UC54 - View Revenue Overview Dashboard		
Created by	Do Duc Anh		
Primary Actor	Manager	Secondary Actor	None
Description	This function allows managers to view overview metrics on the fruit store's business performance, including total revenue, number of orders, growth charts, and AI-powered revenue forecasts for the following month, categorized by time periods (Day/Month/Year).		
Preconditions	1. The manager has successfully logged into the system. 2. The manager has access to the Dashboard & Report module.		
Main Sequence	1. Manager selects the "Dashboard & Report" section from the navigation menu. 2. System retrieves and displays the overview metrics for the current month by default, including total revenue, total successful/canceled orders, and a line chart of daily revenue trends. 3. System requests the AI Module to analyze historical data and displays the AI-powered revenue forecast for the next month. 4. Manager selects a specific time period filter (Day, Month, Year, or Custom Date Range) and clicks the "Apply" button.		

	5. System filters, recalculates the financial data from the database, and updates the dashboard metrics and charts accordingly.
Alternative Sequences	Step 2: If there is no data recorded for the selected time period, the system displays all metrics as 0, shows an empty chart, and prompts a message: "No data available for this period." Step 3: If the AI Module connection fails or times out, the system loads the standard financial charts normally but displays a temporary error message in the forecast section: "AI forecasting is currently unavailable. Please try again later."
Post conditions	System successfully displays the requested business performance and financial data to the manager without modifying any database records.
Business Rule	BR-40

• **3.7.2 UC 55: Analyze Revenue by Product / Category**

ID and Name	UC55 - Analyze Revenue by Product / Category		
Created by	Do Duc Anh		
Primary Actor	Manager	Secondary Actor	None
Description	This function allows managers to view detailed revenue metrics, charts, and breakdown analysis for specific fruit items or product categories, helping them track sales performance across different time periods (Day/Month/Year).		
Preconditions	1. The manager has successfully logged into the system. 2. The manager has access to the Dashboard & Report module.		
Main Sequence	1. Manager selects the "Product & Category Analytics" section from the navigation menu. 2. System retrieves and displays a pie chart showing the revenue distribution by product categories for the current month, along with a ranking table of top-selling fruit items. 3. Manager selects a specific filter (by Product Category, by individual Fruit Item, or by Time Period) and clicks the "Analyze" button. 4. System processes the request, filters the transactional data from the database, and updates the breakdown metrics, bar charts, and growth trends accordingly. 5. Manager selects a specific fruit item to view its sales velocity and real-time inventory turnover rate.		
Alternative Sequences	Step 2: If a product or category has zero sales during the selected period, the system displays the name in the table with a revenue value of 0 and excludes it from the percentage pie chart. Step 3: If the manager inputs invalid custom date parameters (e.g., end date is prior to start date), the system displays an error message: "Invalid date range. Please select again" and retains the previous valid data view.		
Post conditions	System successfully displays the granular product and category sales analysis to the manager without modifying any database records.		
Business Rule	BR-40		

- **3.7.3 UC 56: Generate AI Sales Forecasts**

ID and Name	UC56 - Generate AI Sales Forecasts		
Created by	Do Duc Anh		
Primary Actor	Manager	Secondary Actor	None
Description	This function allows managers to trigger and view AI-driven predictive analysis for the upcoming month. The system processes historical sales data, seasonal trends, and inventory logs to forecast next month's total revenue and recommend optimal stock levels for specific fruit categories.		
Preconditions	1. The manager has successfully logged into the system. 2. The manager has access to the Dashboard & Report module. 3. The system contains at least 3 months of historical sales and inventory data to ensure baseline prediction accuracy.		
Main Sequence	1. Manager navigates to the "AI Forecasting & Predictive Insights" tab within the dashboard. 2. System displays the most recently generated forecast report, including the execution timestamp and accuracy score. 3. Manager clicks the "Run New AI Forecast Analysis" button to process the latest real-time transactional data. 4. System retrieves historical order data, product categories, and current inventory levels from the database, then feeds them into the AI Analytics Module. 5. System generates and displays a predictive report containing: -Projected store revenue for the next month. -Expected demand curve per fruit category. -Smart stock recommendations (e.g., "Increase Mango stock by 15% due to seasonal demand").		
Alternative Sequences	Step 3: If the manager attempts to run a new forecast within 24 hours of the last successful run, the system displays a notification: "Data is already up to date. You can re-run the AI model in [X] hours" to optimize server performance. Step 4: If the historical database contains insufficient data points for a specific newly added fruit item, the system applies a baseline generic category average and flags the item with a note: "Prediction based on category baseline due to limited sales history."		
Post conditions	System successfully updates the predictive analytics dashboard with new forecast metrics and stores the generated report log in the database.		
Business Rule	BR-40		

- **3.7.4 UC 57: Receive Monthly User Activity Reports**

ID and Name	UC57 - Receive Monthly User Activity Reports
Created by	Trần Bảo Lâm

Primary Actor	System Clock / Scheduler (Trigger Event)	Secondary Actor	Administrator
Description	Automated system function that generates a monthly report of user activity statistics (e.g., new registrations, growth rate, and account status distribution) and automatically sends it to the Administrator via email.		
Preconditions	1. The system is running and scheduler service is active. 2. The Administrator has a valid registered email address. 3. The system contains user registration data for the reporting period.		
Main Sequence	1. System Clock triggers the scheduled event at a predefined time (e.g., 00:00 on the first day of each month). 2. System initiates the monthly reporting process in background mode. 3. System retrieves all user accounts created within the previous month. 4. System calculates user activity metrics, including total new registrations, growth rate compared to the previous month, and account status distribution. 5. System generates a formatted monthly activity report. 6. System sends the report via email to the Administrator.		
Alternative Sequences	Step 3: If no new user registrations exist during the reporting period, the system generates a report with zero values and includes a note: "No new user activities recorded this month." The process continues normally. Step 6: If email delivery fails (e.g., mail server downtime), the system logs the error and schedules an automatic retry after a predefined interval.		
Post conditions	A monthly user activity report is generated and sent to the Administrator's email inbox, or queued for retry if delivery fails. .		
Business Rule	BR-40		

● **3.7.5 UC 58: Inspect System Logs**

ID and Name	UC58 - Inspect System Logs		
Created by	Do Duc Anh		
Primary Actor	Admin	Secondary Actor	None
Description	The administrator has successfully logged into the system. The administrator has root/full access privileges to the System Security & Logs module.		
Preconditions	1. The manager has successfully logged into the system. 2. The manager has access to the Dashboard & Report module. 3. The system contains at least 3 months of historical sales and inventory data to ensure baseline prediction accuracy.		
Main Sequence	1. Admin navigates to the "System Logs & Security Monitoring" section from the admin panel.		

	- System retrieves and displays the real-time stream of the latest log entries, including timestamps, log levels (INFO, WARN, ERROR), Actor IDs, IP addresses, and event descriptions. - Admin inputs specific search criteria (e.g., a specific IP address, an Error Code, or a User ID) and selects a time range filter. - System queries the centralized log database, filters the records, and renders the matched log lines on the interface. - Admin selects a specific error log entry to expand and inspect the detailed stack trace and system state at the time of the incident.
Alternative Sequences	Step 2: If the volume of incoming logs causes a system slowdown or high memory usage, the system automatically triggers a pagination mechanism, loading only the 100 most recent entries per page. Step 4: If an IP address triggers a high frequency of failed login attempts or rapid-fire requests within a short window, the system highlights the log entry in red and prompts an action button: "Ban IP/User temporarily".
Post conditions	System displays the requested log data to the admin for security audit purposes without altering any historical log records.
Business Rule	BR-40

3.8 Global Category & Shops Management

- 3.8.1 UC 59: View Global Categories

ID and Name	UC59: View Global Categories		
Created by	Trần Bảo Lâm		
Primary Actor	Admin	Secondary Actor	None
Description	Allows any Customer (including unauthenticated users) to browse the global product category tree and navigate fruit products through hierarchical categories and sub-categories.		
Preconditions	The user is visiting the website (login is not required for browsing).		
Main Sequence	- The Customer hovers over or clicks the “Categories” menu on the navigation bar. - The system retrieves all active main categories and their corresponding sub-categories. - The system displays a structured category dropdown/tree menu. - The Customer selects a specific category or sub-category. - The system filters products based on the selected category. - The system redirects the Customer to the Product Listing page with filtered results.		
Alternative Sequences	Step 2a: No categories available 2a1. The system detects that no categories are configured or active.		

	2a2. The system displays the message: “No categories available at this time.”
Post conditions	The system displays a structured category tree and successfully filters products based on the selected category.
Business Rule	BR-20, BR-39

- **3.8.2 UC 60: Create System Category**

ID and Name	UC60: Create System Category		
Created by	Trần Bảo Lâm		
Primary Actor	Admin	Secondary Actor	None
Description	Allows the Administrator to create a new product category or sub-category in the system, enabling structured organization of fruit products for browsing, filtering, and management.		
Preconditions	1. The Administrator is successfully logged into the system. 2. The Administrator has access to the Category Management module.		
Main Sequence	1. The Administrator navigates to the “Category Management” section and selects “Add New Category”. 2. The system displays the Create Category form, including fields: Category Name, Description, Display Order, and optional Parent Category. 3. The Administrator enters category information and selects a Parent Category if creating a sub-category. 4. The Administrator clicks “Save”. 5. The system validates the input data, ensuring required fields are filled and the category name is unique. 6. The system saves the new category into the database. 7. The system displays a success message: “Category created successfully!” and redirects to the Category List page.		
Alternative Sequences	Step 5a: Duplicate Category Name 5a1. The system detects that the category name already exists. 5a2. The system rejects the request and displays: “This category name already exists. Please choose a unique name.” Step 5b: Missing required fields 5b1. The system detects missing mandatory fields. 5b2. The system highlights invalid fields and displays: “Please fill out all required fields.”		
Post conditions	A new category is successfully created and stored in the system, and becomes available for product assignment and filtering.		
Business Rule	BR-04, BR-07, BR-39, BR-40		

- **3.8.3 UC 61: Update System Category**

ID and Name	UC61: Update System Category		
Created by	Trần Bảo Lâm		
Primary Actor	Admin	Secondary Actor	None

Description	Allows the Administrator to modify existing fruit category information, including category name, description, display order, or parent-child relationship within the category hierarchy.
Preconditions	1. The Administrator is successfully logged into the system. 2. The Administrator is currently viewing the Category List page in the Admin Dashboard. 3. The selected category exists in the system.
Main Sequence	1. Administrator selects a category from the list and clicks “Edit”. 2. System loads the Update Category form with existing category information pre-filled. 3. Administrator updates category details (e.g., name, description, or parent category). 4. Administrator clicks “Save Changes”. 5. System validates updated data, ensuring required fields are filled and rules are satisfied. 6. System saves the updated category information into the database. 7. System displays a success message: “Category updated successfully!” and returns to the Category List page.
Alternative Sequences	Step 5a: Duplicate Category Name 5a1. The system detects that the updated category name already exists in another category. 5a2. The system rejects the update and displays: “This category name is already taken. Please choose another.” Step 5b: Invalid hierarchy (self or circular reference) 5b1. The system detects that the category is assigned as its own parent or creates a circular parent-child relationship. 5b2. The system blocks the update and displays: “Invalid hierarchy selection. A category cannot be a sub-category of itself or its descendants.”
Post conditions	The category information is updated successfully and immediately reflected in both the admin management interface and the customer-facing category tree.
Business Rule	BR-04, BR-39, BR-40

● 3.8.4 UC 62: Delete System Category

ID and Name	UC62: Delete System Category		
Created by	Trần Bảo Lâm		
Primary Actor	Admin	Secondary Actor	None
Description	Allows the Administrator to remove an obsolete or unused fruit category from the system, ensuring that it is no longer available for product assignment or customer browsing.		
Preconditions	1. The Administrator is successfully logged into the system. 2. The Administrator is currently viewing the Category List page in the Admin Dashboard. 3. The selected category exists in the system.		
Main Sequence	1. Administrator selects a category from the list and clicks “Delete”. 2. System displays a confirmation dialog to confirm the deletion action. 3. Administrator clicks “Confirm Delete”. 4. System validates whether the category can be safely deleted (no assigned products and no sub-categories exist). 5. System removes the category from the database and system category tree. 6. System updates the interface and displays a success message: “Category deleted successfully!”.		
Alternative Sequences	Step 3a: Administrator cancels deletion 3a1. Administrator clicks “Cancel” in the confirmation dialog. 3a2. System closes the dialog and no changes are made. Step 4a: Category contains assigned products 4a1. System detects that the category still contains active products. 4a2. System blocks deletion and displays: “Cannot delete category. Please reassign or remove all products first.” Step 4b: Category has sub-categories 4b1. System detects that the category contains child categories. 4b2. System blocks deletion and displays: “Cannot delete parent category. Please remove or reassign sub-categories first.”		
Post conditions	The selected category is permanently removed from the system and is no longer available in both admin management and customer-facing navigation.		
Business Rule	BR-39, BR-40		

● 3.8.5 UC 63: Create Shop

ID and Name	UC63 – Create Shop		
Created by	Đỗ Nguyễn Hùng		
Primary Actor	Administrator	Secondary Actor	None
Description	Allows an Administrator to create a new shop in the system with required information such as shop name, description, address, contact details, and operational status.		

Preconditions	1. The Administrator is logged into the system. 2. The Administrator has access to Shop Management module.
Main Sequence	1. Administrator navigates to the “Shop Management” section and selects “Create Shop”. 2. System displays the shop creation form. 3. Administrator enters shop information (name, description, address, contact details, status). 4. Administrator clicks “Create”. 5. System validates input data and ensures shop name is unique. 6. System creates a new shop record in the database. 7. System displays success message and updates shop list.
Alternative Sequences	Step 5: If required fields are missing, system displays validation error and requests correction. Step 5: If shop name already exists, system rejects creation and displays message: “Shop name already exists”.
Post conditions	A new shop is successfully created and becomes available in the system.
Business Rule	BR-07, BR-08, BR-09

• **3.8.6 UC 64: View Shops List**

ID and Name	UC64 – View Shops List		
Created by	Đỗ Nguyễn Hùng		
Primary Actor	Administrator / Manager	Secondary Actor	None
Description	Allows administrators or managers to view a list of all shops in the system or only shops assigned to them, including basic shop information and status. .		
Preconditions	1. User is logged into the system. 2. User has permission to access Shop Management module.		
Main Sequence	1. User navigates to “Shop Management” section. 2. System retrieves shop list based on user role. 3. System displays list of shops including name, status, and basic information. 4. User views and optionally selects a shop for further actions.		
Alternative Sequences	Step 2: If user is a Manager, system only shows assigned shops. Step 2: If no shops exist, system displays message: “No shops available”.		
Post conditions	Shop list is successfully displayed to the user.		
Business Rule	BR-07, BR-08		

• **3.8.7 UC 65: Update Shop**

ID and Name	UC65 – Update Shop		
Created by	Đỗ Nguyễn Hùng		
Primary Actor	Administrator / Manager	Secondary Actor	None
Description	Allows an authorized user to update existing shop information such as name, description, address, contact details, and status.		
Preconditions	1. User is logged into the system. 2. User has permission to manage the shop. 3. Shop already exists in the system.		
Main Sequence	1. User selects a shop from the shop list. 2. System displays current shop information. 3. User modifies shop details. 4. User clicks “Save Changes”. 5. System validates updated data. 6. System updates shop information in the database. 7. System displays success message.		
Alternative Sequences	Step 5: If updated shop name already exists, system rejects update and displays error. Step 5: If invalid fields are entered, system requests correction.		
Post conditions	Shop information is successfully updated in the system.		
Business Rule	BR-08, BR-09, BR-10		

• **3.8.8 UC 66: Deactivate/Delete Shop**

ID and Name	UC66 – Deactivate/Delete Shop		
Created by	Đỗ Nguyễn Hùng		
Primary Actor	Administrator	Secondary Actor	None
Description	Allows an Administrator to deactivate or remove a shop if it has no active products or related orders.		
Preconditions	1. Administrator is logged in. 2. Shop exists in system. 3. Shop has no active products or pending orders.		
Main Sequence	1. Administrator selects a shop from list. 2. System displays confirmation dialog. 3. Administrator confirms deletion/deactivation. 4. System checks business constraints (products/orders). 5. System deactivates or removes shop record. 6. System updates shop list and displays success message.		

Alternative Sequences	Step 3: If admin cancels, system aborts operation. Step 4: If shop has active products/orders, system blocks deletion and shows error message.
Post conditions	Shop is either deactivated or removed and no longer active in the system.
Business Rule	BR-07, BR-12, BR-13

• **3.8.9 UC 67: Manage Shop**

ID and Name	UC67 – Manage Shop (Operations)		
Created by	Đỗ Nguyễn Hùng		
Primary Actor	Manager	Secondary Actor	None
Description	Allows Manager to manage shop operations including product management, order processing, and inventory control within assigned shop.		
Preconditions	1. Manager is logged in. 2. Manager has selected a shop context.		
Main Sequence	1. Manager selects shop context. 2. System loads shop dashboard. 3. Manager performs operations (product, order, inventory). 4. System processes and updates data. 5. System confirms successful operations.		
Alternative Sequences	Step 2: If no shop selected, system forces selection first.		
Post conditions	Shop data is updated based on manager actions.		
Business Rule	BR-07, BR-10		

• **3.8.10 UC 68: Assign Shop to Manager**

ID and Name	UC68 – Assign Shop to Manager		
Created by	Đỗ Nguyễn Hùng		
Primary Actor	Administrator	Secondary Actor	None
Description	Allows an Administrator to assign one or multiple shops to a manager for operational management.		
Preconditions	1. Administrator is logged in. 2. Manager account exists. 3. Shop exists in system.		
Main Sequence	1. Administrator selects a manager account. 2. System displays available shops. 3. Administrator selects one or more shops. 4. Administrator clicks “Assign”.		

	5. System stores assignment mapping. 6. System confirms successful assignment.
Alternative Sequences	Step 3: If no shop selected, system shows validation error.
Post conditions	Manager is successfully assigned to selected shops.
Business Rule	BR-06, BR-07

4. Non-Functional Requirements

4.1 External Interfaces

4.1.1 User Interfaces (UI)

- The system shall provide a responsive web interface compatible with desktop, tablet, and mobile devices.
- The UI shall be designed for simplicity and ease of use, enabling users to perform core actions (browse products, search, add to cart, checkout) efficiently.
- The system shall provide real-time feedback messages (success, error, validation alerts).

4.1.2 Software Interfaces

- **Frontend:** ReactJS 18
- **Backend:** Spring Boot RESTful API
- **Database:** MySQL
- **Payment Gateway:** VNPAY or Cash on Delivery (COD simulation)
- **AI Module:** Sales forecasting and predictive analytics service
- **Email Service:** SMTP-based email system for automated reports and notifications

4.1.3 Communication Interfaces

- The system shall use RESTful APIs with JSON format for communication between frontend and backend.
- All communication shall be secured using HTTPS protocol.
- The system shall support webhook callbacks for payment gateway confirmation.
- External services (payment, email, AI module) shall communicate through secure API integration.

4.2 Quality Attributes

4.2.1 Usability

- The system shall be easy to use for both technical and non-technical users.
- Core user actions (search, add to cart, checkout) shall require no more than 3–5 steps.
- The UI shall maintain consistency across all modules (e.g Customer and Admin).
- The system shall provide clear validation messages and error notifications.
- Product browsing and filtering shall be intuitive and fast.

4.2.2 Reliability

- The system shall ensure a minimum uptime of 99%.
- All critical operations (orders, payments, inventory updates) must guarantee data consistency.
- The system shall implement rollback mechanisms for failed transactions.
- All important system events shall be logged for auditing and debugging purposes.
- The database shall be backed up periodically to prevent data loss.

4.2.3 Performance

- Average page load time shall be under 4 seconds.
- API response time shall be under 1 second for standard operations.
- The system shall support 500–1000 concurrent users.
- Caching mechanisms shall be applied for frequently accessed data (categories, products).
- AI forecasting processes shall run asynchronously to avoid blocking user interactions.

4.2.4 Security

- The system shall use JWT-based authentication for user login sessions.
- Role-based access control (RBAC) shall be implemented for Customer, Admin, Manager, and Shipper.
- Passwords shall be encrypted using BCrypt hashing algorithm.
- All communications shall be secured via HTTPS.
- Administrative actions shall be logged for audit purposes.
- Sensitive payment information shall not be stored in the system and shall be handled by third-party payment gateways.

5. Requirement Appendix

5.1 Business Rules

ID	Rule Definition
BR-01	Only authenticated users may perform cart management, checkout, order management, refund requests, review submission, address management, and profile modification actions. Product browsing is allowed without login.
BR-02	The system shall support role-based access control (RBAC) for Customer, Administrator, Manager, and Shipper roles.
BR-03	Customers shall only access and modify information belonging to their own account.
BR-04	Required input fields shall not be empty before submission.
BR-05	Email addresses and phone numbers shall follow valid system-defined formats.
BR-06	Password changes require correct current password verification before update.
BR-07	All numeric input fields (e.g., quantity, stock quantity, rating score) shall only accept positive whole numbers. Invalid characters, decimals, and negative values shall be blocked or filtered.
BR-08	Each Customer may maintain multiple delivery addresses.
BR-09	Only one delivery address may be marked as the default address at a time.
BR-10	When the default delivery address is deleted, another available address shall automatically become the default address if one exists.

BR-11	Customers may edit or delete only addresses belonging to their own account.
BR-12	Delivery information (recipient name, phone number, and address details) must be validated before checkout is allowed.
BR-13	Requested cart quantity must not exceed the available stock quantity of the corresponding shop product inventory.
BR-14	Product availability and inventory quantity must always be revalidated during checkout to prevent overselling and race conditions. .
BR-15	The system shall prevent negative inventory quantities under all conditions.
BR-16	Inventory stock shall be temporarily locked during checkout confirmation and online payment processing.
BR-17	Failed or cancelled payment transactions shall automatically release reserved inventory quantities.
BR-18	An order cannot be finalized unless inventory validation and stock deduction are completed successfully in real time.
BR-19	Once an order is confirmed, purchased items, quantities, and unit prices shall become immutable.
BR-20	Inactive or unavailable products shall not be purchasable.
BR-21	Shopping cart totals and sub-totals shall automatically recalculate whenever quantities are updated or products are removed.
BR-22	Shipping fees shall be dynamically calculated per shop order based on shipping method and delivery destination.
BR-23	Every successful order shall generate a corresponding order and invoice/payment record.
BR-24	Orders shall follow the lifecycle: Pending → Confirmed → Shipping → Delivered → Completed / Cancelled.
BR-25	Customers may only access order history, order details, cancellation, review, and refund functions belonging to their own account.
BR-26	Orders in the Order History page shall be displayed in reverse chronological order (newest to oldest).
BR-27	Customers may only cancel eligible orders (e.g., Pending or Preparing status).
BR-28	A cancellation reason shall be selected before an order cancellation request is submitted.
BR-29	Cancelled order quantities shall automatically be restored to warehouse inventory.
BR-30	Only products belonging to completed orders may be reviewed or rated.
BR-31	Rating values shall follow the supported scale (e.g., 1–5 stars).
BR-32	Each purchased product may receive only one review per completed order.
BR-33	Refund or return requests shall only be allowed for eligible completed orders within 7 days.
BR-34	Refund or return requests shall include mandatory information (reason and refund method). Supporting evidence may be required depending on refund type.
BR-35	Duplicate refund requests for the same purchased product/order shall not be allowed.
BR-36	Refund or return requests must be reviewed and approved before refund processing occurs.
BR-37	Notifications (order updates, refunds, payment updates, important account events) shall be delivered in near real-time and immediately reflected in the interface.

BR-38	The unread notification counter shall always reflect the exact number of unread notifications.
BR-39	Category names must be unique and follow a strict parent-child hierarchy without circular relationships (self-parenting or loops).
BR-40	Only Administrators may manage categories, user accounts, shops, and shop assignments..., while system logs shall remain read-only and scheduled reports/analytics shall run automatically according to the official system timezone.

5.2 Common Requirements

CR-01: Validation Rules

- *All user inputs must be validated at both frontend and backend levels.*
- *Error messages must be clear, user-friendly, and consistent.*

CR-02: Data Consistency

- *All transactional operations (order, payment, refund, inventory update) shall ensure ACID properties.*
- *System must prevent partial updates in case of failure.*

CR-03: Session Management

- *Authentication tokens shall expire after a defined period of inactivity or token lifetime.*
- *Sensitive actions (checkout, password change) require re-validation of authentication.*

CR-04: Logging & Monitoring

- *System must log all critical operations including login attempts, admin actions, and financial transactions.*
- *Logs must be timestamped and stored securely.*

CR-05: Error Handling

- *System must handle all exceptions gracefully without crashing.*
- *Users must always receive a meaningful error message instead of system-level errors.*

CR-06: Scalability

- *System must be designed to support future expansion of product categories, users, and order volume.*

CR-07: Maintainability

- *Code structure must follow modular architecture (Controller – Service – Repository pattern).*
- *API endpoints must be clearly documented (Swagger/OpenAPI).*

5.3 Application Messages List

#	Message code	Message Type	Context	Content
1	MSG01	In line	No search results	<i>No results found.</i>
2	MSG02	Validation	Required fields missing	<i>This field is required.</i>
3	MSG03	Toast message	Update success	<i>Updated successfully.</i>
4	MSG04	Toast message	Create success	<i>Created successfully.</i>
5	MSG05	Toast message	Email sent	<i>A confirmation email has been sent to {email_address}.</i>
6	MSG06	Toast message	Delete success	<i>Deleted successfully.</i>
7	MSG07	Toast message	Action success	<i>Operation completed successfully.</i>
8	MSG08	Validation	Input length exceeded	<i>Exceeds maximum allowed length ({max_length}).</i>
9	MSG09	Error	Login failure	<i>Incorrect username or password. Please try again.</i>
10	MSG10	Warning	Stock limit	<i>Requested quantity exceeds available stock.</i>
11	MSG11	Error	Unauthorized access	<i>You do not have permission to perform this action.</i>
12	MSG12	Error	System failure	<i>Something went wrong. Please try again later.</i>
13	MSG13	Info	Empty cart	<i>Your cart is currently empty.</i>
14	MSG14	Warning	Payment failure	<i>Payment failed. Please try again or use another method.</i>
15	MSG15	Success	Order placed	<i>Your order has been placed successfully.</i>